

Air-and-Water Heating System of DENERG



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Chapter 1

Introduction

This document outlines the design and dimensioning of a mixed air-water HVAC system for the section of the Politecnico di Torino building occupied by the Department of Energy (DENERG).

The building consists of five levels: a ground floor and four upper stories, each primarily housing office spaces. Additionally, the facility includes a two-storey laboratory area. For simplification purposes, the second floor is taken as representative of all office levels, allowing the building to be modeled as having five identical office floors and a separate two-level laboratory block.

The main building features office spaces extending across all five floors, while the laboratory structure includes five additional offices distributed across its two storeys. As a preliminary assumption, corridors and stairwells are considered to be at the same temperature as the adjacent office spaces. While this is not strictly accurate—particularly for the stairwells, which experience heat exchange with both the exterior and the laboratory—it serves as a reasonable first-order approximation for the current stage of analysis.

The HVAC system consists of fan coil units in the office areas and a ducted air distribution system in the laboratories. Each office is assumed to contain a single fan coil unit. Mechanical ventilation is not provided in the office zones; instead, air exchange is managed manually through window openings. In contrast, the laboratories are fully air-conditioned with a dedicated air handling system.

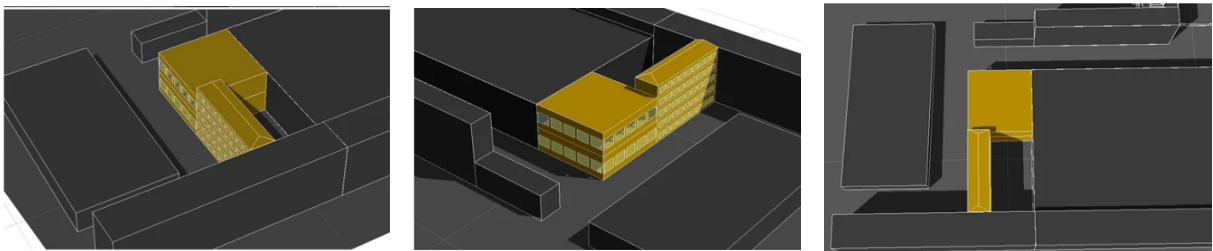


Figure 1.1: 3D Rendering of the DENERG [3]

Furthermore, it is possible to see a second floor plan in figure 1.2 for a better understanding of floor.



Figure 1.2: Second floor plan [3]

Chapter 2

Data collection & preliminary sizing

2.1 Layout of the building

Before being able to determine the thermal load, or any other parameter, it is essential to determine the layout of the building. As previously said, a simplification has been made, namely the fact that the second storey has been considered representative of all 5 storeys present, something that isn't 100% accurate, but good enough for this kind of study.

A dimensioned drawing of the second storey can be seen in figure 2.18:

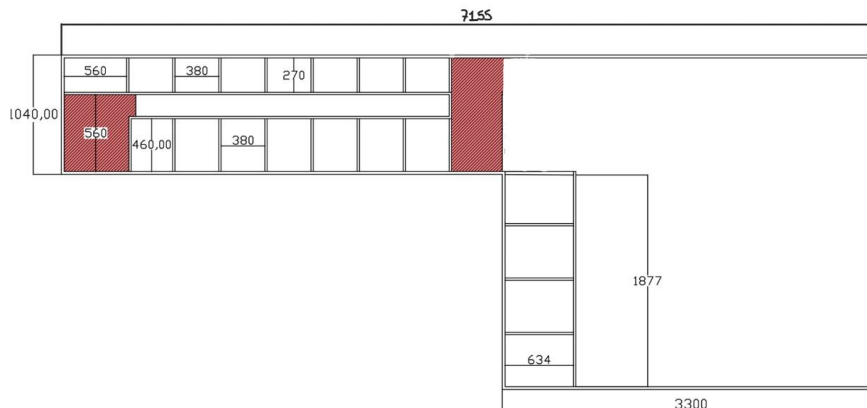


Figure 2.1: Dimensioned drawing of the floor plan

As visible the storey is comprised of 19 offices and 1 laboratory. We can see the main part of the building with just offices (on the left) that has 15 offices, each with a height of 3 meters, and go up 5 storeys. On the right, we have the laboratory and 4 more offices, these offices go up 6 meters and only 2 storeys (the same as the laboratory). Despite the difference in dimensions, the offices are all heated and cooled to the same temperatures, which are different than those used for the laboratory both in summer and winter.

It is important to note, that with our consideration of the layout, we have the first 4

floors, which are all the same essentially, but the fifth office floor does share a wall with the laboratory, but rather all its walls are external.

Another simplifying assumption which was made is that all the windows are the same between the offices and the laboratories. The window considered can be seen in figure 2.2 and was measured directly from an office. It has three transparent surfaces: two identical squares and one rectangle above.

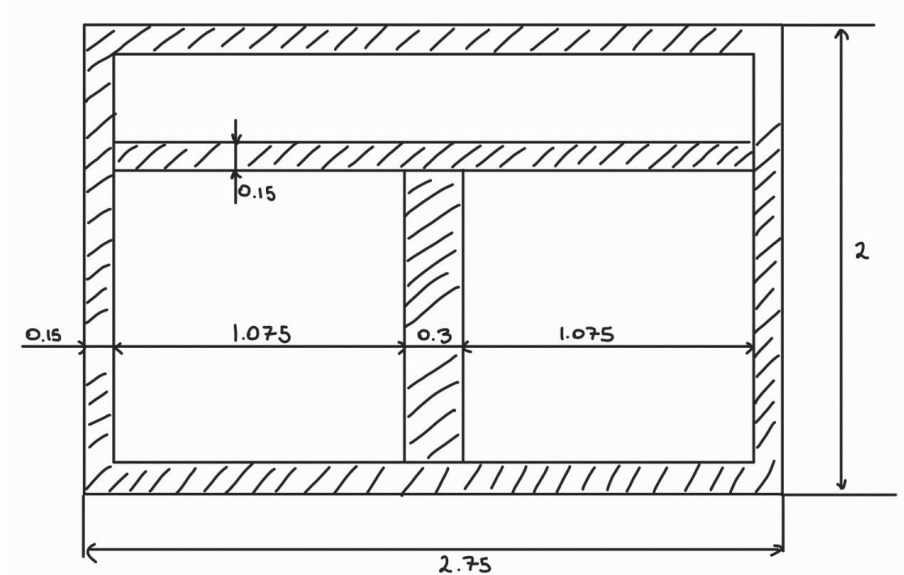


Figure 2.2: Windows of DENERG

With the provided dimensions, it is possible to compute both the total area of the window, and the transparent area of the window, useful for the computation of thermal load due to irradiation in summer:

Total area	5.5	<i>m</i>
Total transparent area	3.475	<i>m</i>

Table 2.1: Surfaces for a single window

Important dimensions used for the computation of the thermal loads are highlighted in Table 2.2

2.2 Thermal Load

Following the computation of the dimensions of the DENERG storeys, it is possible to proceed with the computation of the thermal load. The thermal load has been estimated in a conservative manner, by considering the hottest summers day, and the coldest winter day, thus the two worst conditions.

A further simplification which has been made is to disregard all internal loads, such as electric cables, humans, lights or computers.

RELEVANT DIMENSIONS FOR THERMAL LOAD		
Total wall area between offices and outside	984.72	m^2
Total wall area between lab and outside	735.96	m^2
Total window area (including frame) for offices	660	m^2
Total window area (including frame) for lab	330	m^2
Roof = basement area of offices	519.82	m^2
Roof = basement area of labs	843.61	m^2
Volume of offices	4567.5	m^3
Volume of labs	10123.3	m^3

Table 2.2: (TO BE COMPLETED)

It is possible to start with the computation of the *degree-days* of Turin using the following formula:

$$DD = \sum_{i=1} (T_i - T_e), \quad T_i > T_e \quad (2.1)$$

$$DD_{Turin} = 2617$$

Since Turin has a value of 2617 it is part of thermal zone E (from 2100 - 3000)

The thermal load is estimated by considering a few factors:

1. Heat exchange through external walls
2. Heat exchange through inner walls
3. Heat exchange through windows
4. Heat exchange through the roof
5. Heat exchange through the basement floor
6. Thermal bridges
7. Solar irradiance through the windows

By referencing the same article used for the planimetry and 3D render of the DENERG it is possible to use the already-measured average thermal transmittance of the building, noting that the DENERG building was constructed in 1958, and thus has a non-insulated envelope:

For this project, it has been considered that all the storeys are kept at the same temperature and humidity, therefore at a steady state there is no heat exchanges from one storey to another.

For what regards the ventilation, it was deemed sufficient to consider the following volumetric flow rate:

$$G_{vent} = 2 \cdot \frac{volumes}{hour} \quad (2.2)$$

Component	Transmittance U $\frac{W}{m^2K}$
Walls	1.41
Windows	3.68
Roof	1.30
Basement floor	1.15

Table 2.3: Thermal transmittance estimation of relevant components [3]

Next it is essential to define the internal comfort conditions as well as the external conditions for both winter and summer. We have referred to “*UNI 5364 Italian standards*” for the external comfort conditions, and “*Decreto del presidente della repubblica 16 aprile 2013, n.74*” for the internal comfort conditions.

Typical indoor comfort conditions (office)			
Season	Air temperature	Relative Humidity	Air speed
Winter	20 °C	0.6	0.05 m/s
Summer	25 °C	0.5	0.1 m/s

Table 2.4: Indoor comfort conditions for an office environment.

Typical indoor comfort conditions (lab)			
Season	Air temperature	Relative Humidity	Air speed
Winter	18 °C	0.5	0.05 m/s
Summer	24 °C	0.5	0.1 m/s

Table 2.5: Indoor comfort conditions for a laboratory environment.

2.3 Winter conditions

It is possible to define the formula used for the computation of a thermal load across an opaque surface flowing into a control volume:

$$\Phi_T = \sum_{j=1}^N U_j A_j (T_e - T_i) \quad (2.3)$$

where:

- U_j is the overall transmittance of the surface
- A_j is the area across which the thermal load is transferred
- T_i is the internal temperature
- T_e is the external temperature

External conditions (offices and lab)			
Season	Air temperature	Relative Humidity	Air speed
Winter	-8 °C	0.8	1.7 m/s
Summer	30.5 °C	0.72	0.01 m/s

Table 2.6: Outdoor conditions in Turin

It is important to note that this will provide a positive value if the thermal load is “entering” the building, therefore in winter this value is expected to be negative, while in summer positive.

2.3.1 External heat exchanges

Heat exchange through external walls and offices:

The offices have an external wall all around the main building which goes up five storeys, and there are also 4 offices in the laboratory which go up the equivalent of 4 storeys of the main building (the laboratory goes up 2 storeys, but each storey is twice the height of an office).

By referring to the dimensioned drawing, it is possible to compute the **total external area** (including the windows) between office spaces - including stairwells - and the external environment.

Next we must subtract the area made up by windows. By referring to the drawing in Figure 1.2 we can see that the offices have a total of 120 windows: 20 windows in the main office building per storey, and 5 windows in the offices in the laboratory, which we have for four storeys. We consider all the windows to be the same.

With these values it is possible to compute the total wall area between offices and the external environment:

Total area (including windows) [m^2]	1644.72
Total windows [m^2] - $A_{win,off}$	660
Total wall area [m^2] - $A_{wall,off}$	984.72

Table 2.7: Total wall and windows area between offices and outside

Therefore, it is possible to compute the total thermal load across the external office walls in winter as:

$$\begin{aligned}
 \Phi_{off,wal,wint} &= 1.41 \cdot 984.72 \cdot ((-8) - 20) \\
 &= -38\,877\,W
 \end{aligned}
 \tag{2.4}$$

Heat exchange through windows and offices:

Another source of thermal load for the offices from the outside in winter is the heat exchanged through the windows. For this a similar procedure as before was followed:

- $T_i = 20^\circ C$
- $T_e = -8^\circ C$
- $U = 3.68 \frac{W}{m^2 K}$

As computed previously, the total area made up by windows between the offices and the outside is

$$A_{win,off} = 660 \text{ m}^2$$

By using equation 2.3 we can compute the thermal load transmitted through the office windows in winter:

$$\begin{aligned} \Phi_{off,win,wint} &= 3.68 \cdot 660 \cdot ((-8) - 22) \\ &= -68\,006 \text{ W} \end{aligned} \tag{2.5}$$

Heat exchange through external walls and lab

As for the lab, according to the already mentioned information, we have an external perimeter of 88.83 m, a height of 6 m, for two storeys and 15 windows of the same dimensions as those in the offices. The windows are considered twice per storey.

Total area (including windows) $[m^2]$	1065.96
Total windows $[m^2]$ - $A_{win,lab}$	330
Total wall area $[m^2]$ - $A_{wall,lab}$	735.96

Table 2.8: Total wall and windows area between the laboratory and outside

Furthermore, it is important to recall that the internal office temperature is set at $18^\circ C$ instead of $20^\circ C$, while the external temperature is the same as in the previous computations.

With all this, it is possible to compute the thermal load across the external laboratory wall in winter:

$$\begin{aligned} \Phi_{lab,wall,wint} &= 1.41 \cdot 735.96 \cdot ((-8) - 18) \\ &= -26\,980 \text{ W} \end{aligned} \tag{2.6}$$

Heat exchange through windows and lab

This heat exchange was computed in a very similar fashion as before:

- $T_i = 18^\circ C$
- $T_e = -8^\circ C$
- $U = 3.68 \frac{W}{m^2 K}$
- $A_{win,lab} = 330 m^2$

$$\begin{aligned}\Phi_{lab,win,wint} &= 3.68 \cdot 330 \cdot ((-8) - 18) \\ &= -31\,574\,W\end{aligned}\tag{2.7}$$

Heat exchanged through the roof and basement

As already discussed, there is no heat exchanged through the floors from one storey to the next, however there is heat which is exchanged through the roof and the basement. For what regards the heat exchanged through the roof we can consider first the roof above the offices:

$$A_{roof,off} = 519.8\,m^2$$

$$\begin{aligned}\Phi_{roof,off,wint} &= 1.30 \cdot 519.8 \cdot ((-8) - 20) \\ &= -18\,921\,W\end{aligned}\tag{2.8}$$

Whereas for what regards the roof above the laboratory we have:

$$A_{roof,lab} = 843.6$$

$$\begin{aligned}\Phi_{roof,lab,wint} &= 1.30 \cdot 843.6 \cdot ((-8) - 18) \\ &= -28\,514\,W\end{aligned}\tag{2.9}$$

There is heat also exchanged through the basement. We can consider the basement areas to be the same as the roof areas. The transmittance is specified in table 2.3, while the external temperature can be found looking at the Figure 2.3

By considering the depth of the basement as $2\,m$ and by considering the worst case, the external temperature is $T_e = 5^\circ C$.

With these considerations we can compute the heat transfer across the basement:

$$\begin{aligned}\Phi_{bas,off,wint} &= 1.15 \cdot 519.8 \cdot (5 - 20) = -8\,967\,W \\ \Phi_{bas,lab,wint} &= 1.15 \cdot 843.6 \cdot (5 - 18) = -12\,612\,W\end{aligned}\tag{2.10}$$

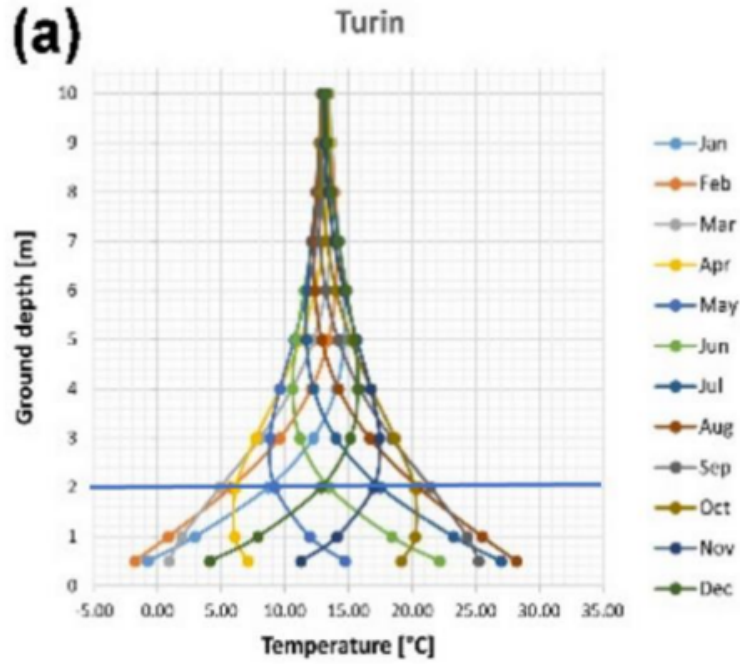


Figure 2.3: Basement temperatures in Turin

2.3.2 Internal heat exchanges

Inside, there will be heat exchanges due to the fact that the offices and laboratory are kept at two different internal temperatures. As already mentioned, the heat exchanges at the stairwells are neglected at the moment as they are considered to be the same temperature as the offices.

The internal heat exchanged is computed by considering that the transmittance of the walls between the offices and the laboratory is the same as the external walls, and that it is the same as the transmittance between the stairwell and the laboratory.

By referring Figure 2.18 it is possible to see that the perimeter between the offices and laboratory is 35.31 m, and this spans across both laboratory storeys, resulting in a total area:

$$\begin{aligned} A_{off,lab} &= (35.31 \cdot 6) \cdot 2 \\ &= 426.12 \text{ m}^2 \end{aligned} \quad (2.11)$$

With this value it is possible to compute the thermal load going **from the offices to the laboratory**:

$$\begin{aligned} \Phi_{int,lab,wint} &= 1.41 \cdot 426.12 \cdot (20 - 18) \\ &= 1202 \text{ W} \end{aligned} \quad (2.12)$$

Since this is an internal load, it is perfectly balanced by an equal and opposite load

going from the laboratory to the office:

$$\Phi_{int,off,wint} = -1\,202\,W \quad (2.13)$$

2.3.3 Consideration of thermal bridges

With all the above computations it is possible to summarize them in the following two tables:

Offices Thermal loads	Value [W]
$\Phi_{ext,wall}$	-38 877
Φ_{int}	-1 202
$\Phi_{ext,win}$	-68 006
Φ_{roof}	-18 921
Φ_{bas}	-8 967
$\Phi_{off,wint}$	-135 973

Table 2.9: Thermal Loads in Offices

Laboratory Thermal loads	Value [W]
$\Phi_{ext,wall}$	-26 980
Φ_{int}	1 202
$\Phi_{ext,win}$	-31 574
Φ_{roof}	-28 514
Φ_{bas}	-12 612
$\Phi_{lab,wint}$	-98 479

Table 2.10: Thermal Loads in the laboratory

Finally, according to the regulations outlined in UNI TS 11300-1 2008, a correction factor of 10% is applied to accommodate for imperfect insulation and other possible thermal bridges. Due to this correction factor we can get the final thermal loads for the winter months:

The total thermal load for the offices excluding the ventilation load:

$$\Phi_{off,wint} = 1.1 \cdot (-135\,973) = -149\,570\,W \quad (2.14)$$

While for the laboratory it is

$$\Phi_{lab,wint} = 1.1 \cdot (-98\,479) = -108\,327\,W \quad (2.15)$$

2.3.4 Heat exchanges due to ventilation

As already mentioned, the ventilation in the offices will be manual, while in the lab it will be done with the air circulation system. The air volumetric flow rate required is the following:

$$G_{vent} = 2 \cdot \frac{volumes}{hour} \quad (2.16)$$

where the volume considered is the volume of just the offices, excluding the corridors and the stairwells. This volume is $V_{off} = 4567.5 m^3$.

With this ventilation, there is a heat load associated to it:

$$\Phi_{vent,off} = \rho_{air} \cdot G_{vent} \cdot c_p \cdot (T_e - T_i) \quad (2.17)$$

Considering air as an ideal gas (with $R = 287 \frac{J}{kg \cdot K}$) and by using the external temperature to compute ρ_{air} , and $c_p = 1004.5 \frac{J}{kg \cdot K}$ we can solve the above equation:

$$\begin{aligned} \Phi_{vent,off} &= \frac{p_{atm}}{R \cdot T_e} \cdot 2 \cdot \frac{V_{off}}{3600} \cdot c_p \cdot (T_e - T_i) \\ &= \frac{101325}{287 \cdot 265} \cdot 2 \cdot \frac{4567.5}{3600} \cdot 1004.5 \cdot ((-8) - 20) \\ &= -95030 W \end{aligned} \quad (2.18)$$

For the laboratories a similar procedure was followed, however, since the laboratories are extremely large in volume, and we don't expect a lot of people to be inside of them normally, we considered it appropriate to recirculate 0.5 volumes per hour instead of 2. With this consideration the procedure is analogous to that for the offices:

$$\begin{aligned} \Phi_{vent,lab} &= \frac{p_{atm}}{R \cdot T_e} \cdot \frac{1}{2} \frac{V_{lab}}{3600} \cdot c_p \cdot (T_e - T_i) \\ &= \frac{101325}{287 \cdot 265} \cdot \frac{1}{2} \frac{10123.3}{3600} \cdot 1004.5 \cdot ((-8) - 18) \\ &= -48894 W \end{aligned} \quad (2.19)$$

Finally, we can sum up all the contributions and summarize the total thermal load for the DENERG in winter:

Total thermal load for the offices [W]	$\Phi_{off,wint,tot}$	-244 600
Total thermal load for the laboratory [W]	$\Phi_{lab,wint,tot}$	-157 221
Total thermal load in winter DENERG [W]	Φ_{winter}	401 821

Table 2.11: Total thermal loads for DENERG in winter

2.4 Summer conditions

2.4.1 Heat exchanges common with winter case

In summer, the procedure done is exactly the same, the main difference is that we consider irradiation in summer. In order to not be repetitive we have decided to not reshew every formula used for the thermal loads already computed in winter as the areas are the same. The only difference is the external and internal temperatures, which are:

- External temperature: $T_e = 30.5^\circ C$
- External basement temperature (worst condition in Figure 2.3): $T_{e,base} = 22^\circ C$
- Internal office temperature: $T_{i,off} = 25^\circ C$
- Internal lab temperature: $T_{i,lab} = 24^\circ C$

Other than these differences the procedure is the same, so we can summarize the total thermal load, excluding ventilation and irradiation, in summer for both the labs and offices (already having considered a 10% correction accounting for the thermal bridges):

Φ_{off}	24 549	W
Φ_{lab}	22 471	W

Table 2.12: Thermal loads in summer BEFORE considering ventilation and irradiation

2.4.2 Heat exchanges due to ventilation

The volumetric flow rate considered is the same as in summer, and so are the formulas used, the main difference is that considering a different external temperature we have a different density: For the offices:

$$\begin{aligned}
 \Phi_{vent,off} &= \frac{p_{atm}}{R \cdot T_e} \cdot 2 \cdot \frac{V_{off}}{3600} \cdot c_p \cdot (T_e - T_i) \\
 &= \frac{101\,325}{287 \cdot 303.5} \cdot 2 \cdot \frac{4567.5}{3600} \cdot 1004.5 \cdot (30.5 - 25) \\
 &= -16\,300\,W
 \end{aligned} \tag{2.20}$$

For the labs:

$$\begin{aligned}
 \Phi_{vent,lab} &= \frac{p_{atm}}{R \cdot T_e} \cdot \frac{1}{2} \frac{V_{lab}}{3600} \cdot c_p \cdot (T_e - T_i) \\
 &= \frac{101\,325}{287 \cdot 303.5} \cdot \frac{1}{2} \frac{10\,123.3}{3600} \cdot 1004.5 \cdot (30.5 - 24) \\
 &= -10\,674\,W
 \end{aligned} \tag{2.21}$$

2.4.3 Heat exchanges due to irradiation

Irradiation is a type of heat transfer that occurs by light (electromagnetic waves) hits an object/person. For this reason, in our context, irradiation occurs only through the transparent surfaces, namely the glass part of the windows (we have to subtract the frame from the window total area):

$$A_{wind,glass} = 3.475 \text{ m}^2$$

Furthermore we have to find the irradiation that goes through the windows. To this end we used the online software PVGIS: https://joint-research-centre.ec.europa.eu/photovoltaic-geographical-information-system-pvgis_en.

We consider that the building has light come in from four different *Azimuth* angles, which are measured from north:

- *Azimuth 1*: 23.13°
- *Azimuth 2*: 113.13°
- *Azimuth 3*: -156.85°
- *Azimuth 4*: -66.86°

and obviously an inclination angle of 90° was used since the walls are vertical.

When using the online software, we got values of irradiation for every month, we decided to use the highest value for the summer months, which was July for all angles expect for *Azimuth 1* which was August:

SOLAR IRRADIATION		
<i>Azimuth 1</i>	138.45	$\frac{W}{m^2}$
<i>Azimuth 2</i>	123.16	$\frac{W}{m^2}$
<i>Azimuth 3</i>	76.37	$\frac{W}{m^2}$
<i>Azimuth 4</i>	157.70	$\frac{W}{m^2}$

Table 2.13: Solar irradiation worst conditions of the year

Finally we can compute the thermal load due to irradiation:

Azimuth angle ($^\circ$)	23.13	113.13	-156.85	-66.86
N of Windows (Offices)	45	0	50	24
N of Windows (Laboratory)	48	48	40	0
Irradiance (worst case) ($\frac{W}{m^2}$)	138.45	123.16	76.37	157.70
Irradiation load (offices) [W]	21 651	0	13 270	13 152
Irradiation load (labs) [W]	23 094	21 211	10 616	0

Table 2.14: Azimuth Angles, Windows, and Irradiance Calculations

By summing up all the contributions, we can compute the total thermal load in summer:

Total thermal load for the offices [W]	$\Phi_{off,sum,tot}$	88 921
Total thermal load for the laboratory [W]	$\Phi_{lab,sum,tot}$	88 065
Total thermal load in winter DENERG [W]	Φ_{summer}	176 986

Table 2.15: Total thermal loads for DENERG in summer

2.5 Most critical floor

To size some of the components, for example, the HVAC system, we wanted to compute the most critical storey (so as to have to size the HVAC system only for that storey and consider all others to be the same). To this end, we performed the previous calculations only for the first, fourth, and fifth storeys as the most critical has to be one of these. We arrived at this conclusion since these three storeys have exactly the same contributions as storeys 2 and 3, and other contributions, namely from the basement, roof, or another vertical wall in the case of the fifth storey of the offices (where before there was the lab, now there is an external wall):

Storey	Load	Unit
Floor 1: Offices	-63 712	W
Floor 1: Laboratory	-65 920.3	W
Floor 4: Offices	-58 614	W
Floor 4: Laboratory	-83 413	W
Floor 5: Offices only	-67 195	W

Table 2.16: Winter thermal loads for floors 1, 4 and 5

Storey	Load	Unit
Floor 1: Offices	19 355	W
Floor 1: Laboratory	24 328	W
Floor 4: Offices	21 838	W
Floor 4: Laboratory	34 304	W
Floor 5: Offices only	24 409	W

Table 2.17: Summer thermal loads for floors 1, 4 and 5

It is clear that the most critical laboratory floor is the 4th floor (which is actually the second floor of the laboratory), and the fifth floor for the offices.

2.3.1 HVAC System – Winter Condition

The heating system is made up by:

- An HVAC system for the laboratories (air-based)
- A fan-coil system for the offices (water-based)

The HVAC system layout is illustrated in Figure 2.4.

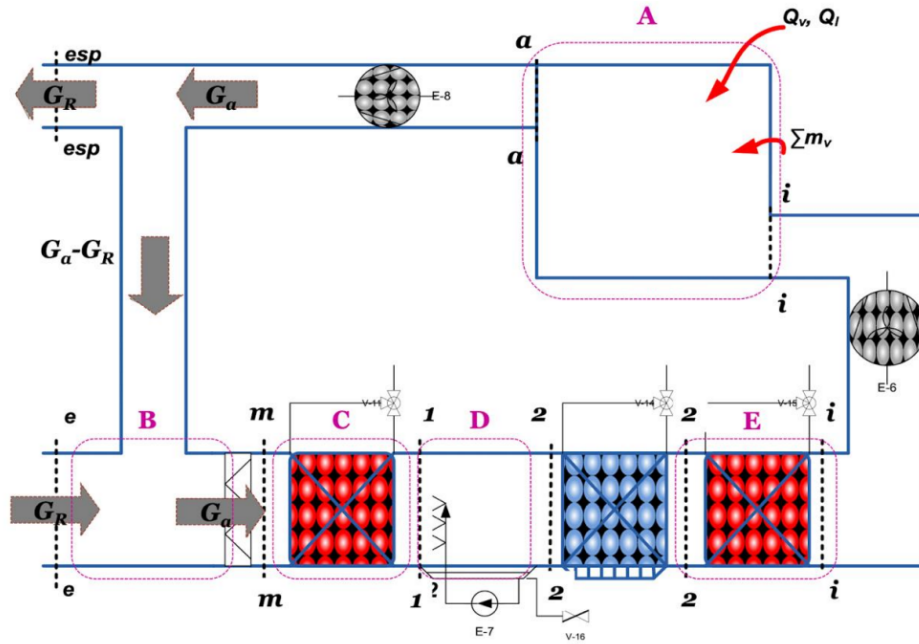


Figure 2.4: HVAC system

In winter conditions, the use of the cooler and post heater is not required in the HVAC system considered.

Design Parameters:

- External Temperature: $T_e = -8^\circ\text{C}$
- Internal Temperature (laboratories): $T_a = 18^\circ\text{C}$
- Relative Humidity in laboratories: $\varphi_a = 50\%$
- Total thermal load 2nd storey: $\Phi_{lab,winter} = -83\,413\text{ W}$

Air refreshment is a critical aspect of maintaining healthy conditions. As previously mentioned, we assumed an air refreshment rate of 0.5 volumes per hour. The standards recommend 2 volumes per hour, however, since the laboratories are very big (6 meters high), and it isn't expected to have a lot of people in them at once, to reduce significantly the thermal load, and the mass flow rate (thus the pressure losses) we decided to consider this reduced air refreshment rate.

Furthermore, to size the HVAC, we considered only the most critical of the two storeys (the one with the highest thermal load), and sized both storeys the same way. The most critical storey is the second one.

Volume of a single laboratory storey:

$$V_{lab} = 5\,691.65\text{ m}^3$$

Volumetric flow rate of refreshed air:

$$G_r = \frac{0.5 \cdot V_{lab}}{3600} = \frac{0.5 \cdot 5\,061.65}{3600} = 0.70 \frac{m^3}{s} \quad (2.22)$$

To compute the mass flow rate we must recall that $\dot{G} = \rho \cdot G$, where ρ is the external air density:

$$\rho_e = \frac{p_e}{R \cdot T_e} = \frac{101\,325}{287 \cdot 265} = 1.33 \frac{kg}{m^3}$$

With these two values it is possible to compute the mass flow rate of the refreshment air:

$$\dot{G}_R = G_r \cdot \rho_e = 0.70 \cdot 1.33 = 0.94 \frac{kg}{s} \quad (2.23)$$

Control Volume A

This control volume considers the laboratory, with inlet i and outlet a .

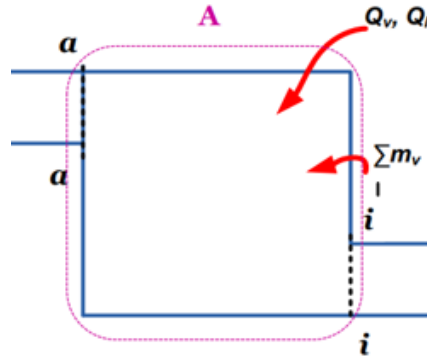


Figure 2.5: Control Volume A

Using the First Law of Thermodynamics for an open system and neglecting both irradiance and latent heat:

$$\Phi_s = \dot{G}_a(h_a - h_i)$$

This can be rewritten to find the necessary air mass flow rate:

$$\dot{G}_a = \frac{\Phi_s}{h_a - h_i} \quad (2.24)$$

Given sensible load $\Phi_s = -83.41 \text{ kW}$.

State 'a':

Humidity ratio:

$$x_a = \frac{0.622 \cdot \varphi_a \cdot p_{a,sat}}{p - \varphi_a \cdot p_{a,sat}} = \frac{0.622 \cdot 0.5 \cdot 0.0206}{1 - 0.5 \cdot 0.0206} = 0.00647 \frac{kg_{vap}}{kg_{dry}} \quad (2.25)$$

It is also possible to define the following three parameters:

- $c_{pa} = 1 \frac{kJ}{kg \cdot ^\circ C}$ is the specific heat capacity of the dry air at constant pressure
- $c_{pv} = 1.0045 \frac{kJ}{kg \cdot ^\circ C}$ is the specific heat capacity of the vapour at constant pressure
- $r_0 = 2500 \frac{kJ}{kg}$ is the latent heat of vaporization

Given these parameters, it is possible to solve for the enthalpy in point 'a':

$$h_a = c_{pa}T_a + x_a(r_0 + c_{pv}T_a) = 1 \cdot 18 + 0.00647 \cdot (2500 + 1.9 \cdot 18) = 34.4 \frac{kJ}{kg} \quad (2.26)$$

State 'i':

The humidity ratio at point i is the same as at point a, therefore:

$$x_i = x_a = 0.00647 \frac{kg_{vap}}{kg_{dry}} \quad (2.27)$$

For what regards h_i , we need to decide the temperature at the inlet of the room. It was decided to start with an initial inlet temperature $T_i = 32^\circ C$. This high temperature was deemed appropriate for two reasons: Firstly out of necessity since the volume of the laboratory is very big, a higher temperature allows for a reduced mass flow rate which reduces the pressure losses; and the other reason is because the ducts would be located near the ceiling, so that the hot air will homogenize with the cold air before reaching the humans, thus no discomfort would be felt.

$$h_i = 1 \cdot 32 + 0.00647 \cdot (2500 + 1.9 \cdot 35) = 48.60 \frac{kJ}{kg} \quad (2.28)$$

It is now possible to calculate an initial air mass flow rate:

$$\dot{G}_a = \frac{\Phi_s}{h_a - h_i} = \frac{-83.41}{34.4 - 48.60} = 5.89 \frac{kg}{s} \quad (2.29)$$

Control Volume B

In this control volume we have the mixing of the external air with the recirculated air. We are able to write the two following balance equations:

$$\begin{cases} \dot{G}_R h_e + (\dot{G}_a - \dot{G}_R) h_a = \dot{G}_a h_m \\ \dot{G}_R x_e + (\dot{G}_a - \dot{G}_R) x_a = \dot{G}_a x_m \end{cases} \quad (2.30)$$

By considering the external relative humidity: $\varphi_e = 80\%$, we can compute the vapour fraction and the enthalpy at point e:

$$x_e = \frac{0.622 \cdot \varphi_e \cdot p_{e,sat}}{p - \varphi_e \cdot p_{e,sat}} = \frac{0.622 \cdot 0.8 \cdot 0.0033}{1 - 0.8 \cdot 0.0033} = 0.00165 \frac{kg_{vap}}{kg_{dry}} \quad (2.31)$$

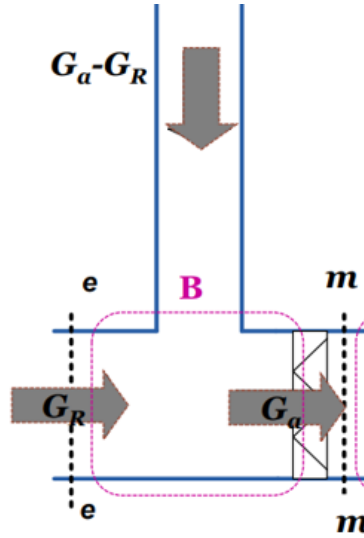


Figure 2.6: Control Volume B

$$h_e = 1 \cdot (-8) + 0.00165 \cdot (2500 + 1.9 \cdot (-8)) = -3.91 \frac{kJ}{kg} \quad (2.32)$$

Now, by rearranging the previous balance equations, we get the mixed enthalpy and vapor fraction at point m :

$$\begin{aligned} h_m &= \frac{\dot{G}_R h_e + (\dot{G}_a - \dot{G}_R) h_a}{\dot{G}_a} \\ &= \frac{0.70 \cdot (-3.91) + (5.89 - 0.70) \cdot 34.4}{5.89} = 28.33 \frac{kJ}{kg} \end{aligned} \quad (2.33)$$

$$\begin{aligned} x_m &= \frac{\dot{G}_R x_e + (\dot{G}_a - \dot{G}_R) x_a}{\dot{G}_a} \\ &= \frac{0.70 \cdot (0.00165) + (5.89 - 0.70) \cdot 0.00648}{5.89} = 0.00571 \frac{kg_{vap}}{kg_{dry air}} \end{aligned} \quad (2.34)$$

Control Volume C (Pre-heater)

After this, the mixed air enters the pre-heater, which heats the air before sending it to the humidifier. The heating occurs at an iso-vapour fraction process, therefore:

$$x_1 = x_m = 0.00571 \frac{kg_{vap}}{kg_{dry}} \quad (2.35)$$

while reducing the relative humidity, and increasing both the temperature and the enthalpy.

Control Volume D (Humidifier)

The humidifier is placed between the pre-heater and the inlet ducts of the room.

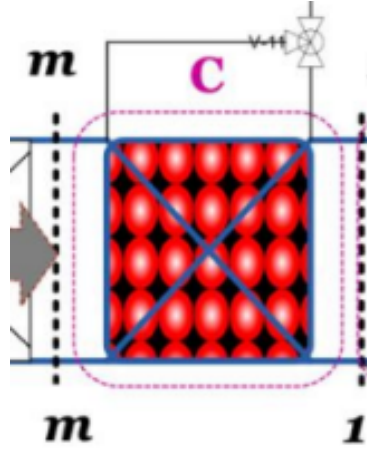


Figure 2.7: Pre-heater

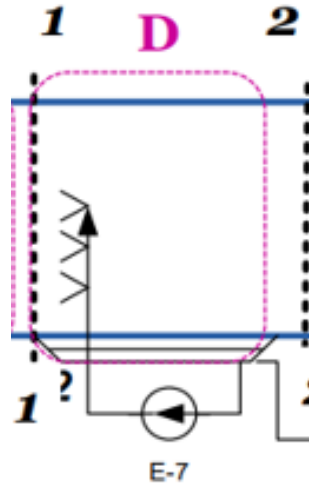


Figure 2.8: Humidifier

In the humidifier we add a certain amount of vapour so as to increase the vapour fraction of the air, while keeping the temperature constant:

Water vapor mass flow:

$$\dot{G}_{vap} = \dot{G}_a(x_2 - x_1) = 5.89 \cdot (0.00648 - 0.00571) = 0.0453 \frac{kg}{s} \quad (2.36)$$

Furthermore, we can write the following two balances:

$$\begin{cases} \dot{G}_a h_1 + \dot{G}_{vap} h_{vap} = \dot{G}_a h_2 \\ \dot{G}_a x_1 + \dot{G}_{vap} = \dot{G}_a x_2 \end{cases} \quad (2.37)$$

By combining the two equations, we get the change in enthalpy:

$$h_2 - h_1 = h_{vap}(x_2 - x_1) = 2675 \cdot (0.00648 - 0.00571) = 2.06 \frac{kJ}{kg} \quad (2.38)$$

Finally we can compute the total power required by the HVAC in winter:

Total power required:

$$\begin{aligned}
 \Phi_C &= \dot{G}_a[(h_1 - h_m) + (h_i - h_2)] \\
 &= \dot{G}_a[(h_1 - h_2) + (h_i - h_m)] \\
 &= 5.89[(2.06) + (48.60 - 28.33)] \\
 &= 131.40 \text{ kW}
 \end{aligned} \tag{2.39}$$

Note: We will see that in summer we have to have a post-heater and don't need a pre-heater. For winter we could have used either only a pre-heater or post-heater, or a combination of the two. We decided to use only a pre-heater since it was stated in the project requirements that the HVAC system has a pre-heater, but we could have chosen to have only a post-heater eliminating the need of a pre-heater in general for this HVAC system.

The HVAC process is also represented on the psychrometric chart:

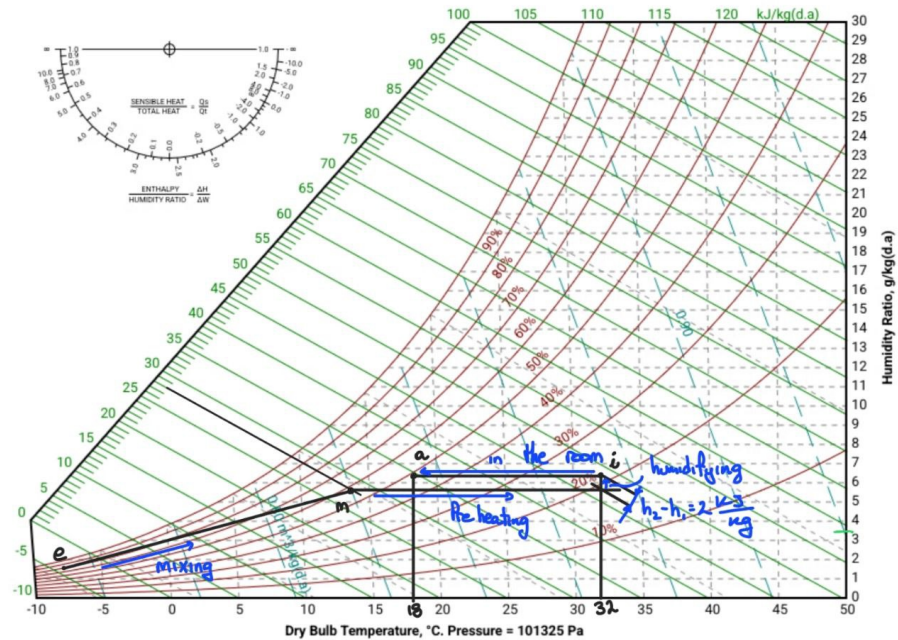


Figure 2.9: HVAC Process for winter

2.6 HVAC System - Summer Conditions

The HVAC system considered in summer is the same as in winter. The design parameters are the following:

- External Temperature: $T_e = 30.5^\circ C$
- Internal Temperature: $T_a = 24^\circ C$
- Internal Relative Humidity: $\varphi_a = 50\%$
- External Relative Humidity: $\varphi_e = 70\%$
- Total thermal load 1st storey: $\phi_s + \phi_l + \phi_{irr} = 34.3 kW$

Again, as in the winter case, we are only sizing the second storey to be conservative. Furthermore, in the summer we don't need to consider the humidifier and the pre-heater, however, the cooler and post-heater will need to be considered.

2.6.1 Control Volume A

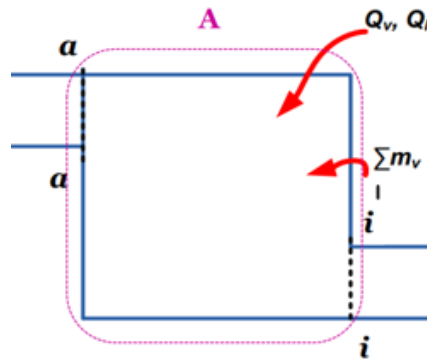


Figure 2.10: Control Volume A

Here, the procedure is very similar to the winter case, with the main difference the fact that we have to consider irradiance as well. It would be more appropriate to also consider a latent load, however, as a preliminary assumption we considered this load to be zero.

Using the First Law of Thermodynamics for an open system:

$$\Phi_s = \dot{G}_a(h_a - h_i)$$

$$\dot{G}_a = \frac{\Phi_s + \Phi_{irr}}{h_a - h_i} \quad (2.40)$$

Given sensible load $\Phi_s = 20.57 \text{ kW}$ and irradiance load $\Phi_{irr} = 13.73 \text{ kW}$.

State 'a':

Humidity ratio:

$$x_a = \frac{0.622 \cdot \varphi_a \cdot p_{a,sat}}{p - \varphi_a \cdot p_{a,sat}} = \frac{0.622 \cdot 0.5 \cdot 0.0298}{1 - 0.5 \cdot 0.0298} = 0.00941 \frac{\text{kg}_{vap}}{\text{kg}_{dry}} \quad (2.41)$$

Next, it is possible to solve for the enthalpy in point 'a':

$$h_a = c_{pa}T_a + x_a(r_0 + c_{pv}T_a) = 1 \cdot 24 + 0.00941 \cdot (2500 + 1.9 \cdot 24) = 47.96 \frac{\text{kJ}}{\text{kg}} \quad (2.42)$$

State 'i':

The humidity ratio at point i is the same as at point a, therefore:

$$x_i = x_a = 0.00941 \frac{\text{kg}_{vap}}{\text{kg}_{dry}} \quad (2.43)$$

For what regards h_i , following a similar reasoning to the choice of T_i in the winter case, we decided to have a very low temperature as the inlet temperature (15°C) so as to minimize the mass flow rate required. Again, since the inlets in the room are in the ceiling, and the laboratory is 6 meters high we think that this value is appropriate and would not create discomfort for the users.

$$h_i = 1 \cdot 15 + 0.00941 \cdot (2500 + 1.9 \cdot 15) = 38.80 \frac{\text{kJ}}{\text{kg}} \quad (2.44)$$

It is now possible to calculate an initial air mass flow rate:

$$\dot{G}_a = \frac{\Phi_s}{h_a - h_i} = \frac{34.30}{47.96 - 38.80} = 3.74 \frac{\text{kg}}{\text{s}} \quad (2.45)$$

2.6.2 Control Volume B

The mixing of the refreshment air and recirculation air is analogous to the winter case:

$$\begin{cases} \dot{G}_R h_e + (\dot{G}_a - \dot{G}_R) h_a = \dot{G}_a h_m \\ \dot{G}_R x_e + (\dot{G}_a - \dot{G}_R) x_a = \dot{G}_a x_m \end{cases} \quad (2.46)$$

By considering the external relative humidity: $\varphi_e = 70\%$ we can compute the vapour fraction and the enthalpy at point e:

$$x_e = \frac{0.622 \cdot \varphi_e \cdot p_{e,sat}}{p - \varphi_e \cdot p_{e,sat}} = \frac{0.622 \cdot 0.7 \cdot 0.0437}{1 - 0.7 \cdot 0.0437} = 0.01961 \frac{\text{kg}_{vap}}{\text{kg}_{dry}} \quad (2.47)$$

$$h_e = 1 \cdot (30.5) + 0.01961 \cdot (2500 + 1.9 \cdot (30.5)) = 80.67 \frac{kJ}{kg} \quad (2.48)$$

Now, by rearranging the previous balance equations, we get the mixed enthalpy and humidity at point m :

$$\begin{aligned} h_m &= \frac{\dot{G}_R h_e + (\dot{G}_a - \dot{G}_R) h_a}{\dot{G}_a} \\ &= \frac{0.80 \cdot (80.67) + (3.74 - 0.80) \cdot 47.96}{3.74} = 54.91 \frac{kJ}{kg} \end{aligned} \quad (2.49)$$

$$\begin{aligned} x_m &= \frac{\dot{G}_R x_e + (\dot{G}_a - \dot{G}_R) x_a}{\dot{G}_a} \\ &= \frac{0.80 \cdot (0.01961) + (3.74 - 0.80) \cdot 0.00941}{3.74} = 0.0116 \frac{kg_{vap}}{kg_{dry air}} \end{aligned} \quad (2.50)$$

2.6.3 Control Volume Cooler

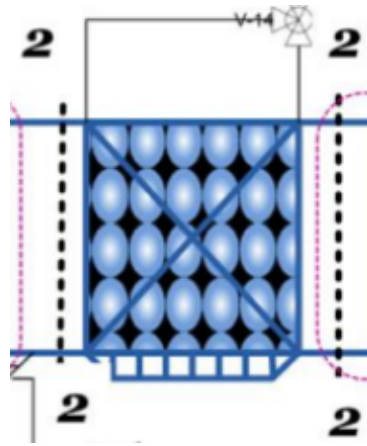


Figure 2.11: Cooler CV

By considering 2 the inlet and 2' the outlet, in the cooler we start by performing an iso-vapor fraction cooling of the air until it reaches saturation. Once the fluid is saturated ($\varphi = 100\%$), if we continue to cool the fluid we cool along the saturation line, thus decreasing the vapor-fraction. We continue the cooling process until the vapor fraction desired at the inlet of the laboratory $x_i = 0.00941$.

Therefore the enthalpy at point $h_{2'}$ is found by imposing: $x_{2'} = x_1$ and $\varphi_{2'} = 100\%$:

$$h_{2'} = 35.60$$

Now we can compute the power required by the cooler:

$$\Phi_{cooler} = \dot{G}_a (h_{2'} - h_2) = -68.90 \text{ kW} \quad (2.51)$$

2.6.4 Control Volume E (Post-heater)

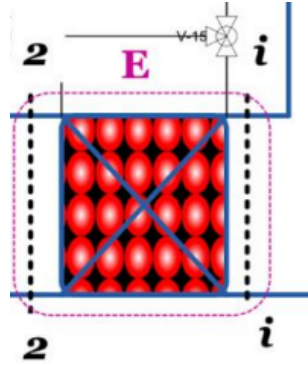


Figure 2.12: Post-heater control volume

In the post-heater, we are performing a very small heating process to increase the temperature of the air to the inlet desired temperature of $T_i = 15^\circ\text{C}$. This is an iso-vapor fraction process: In the post heater, we get the air at state $2'$ and deliver it at state i . Since we know everything about these two states we can compute the power required by this cell:

$$\Phi_{\text{post-heater}} = \dot{G}_a(h_i - h_{2'}) = 8.62 \text{ kW} \quad (2.52)$$

All these processes can also be represented on the psychrometric chart:

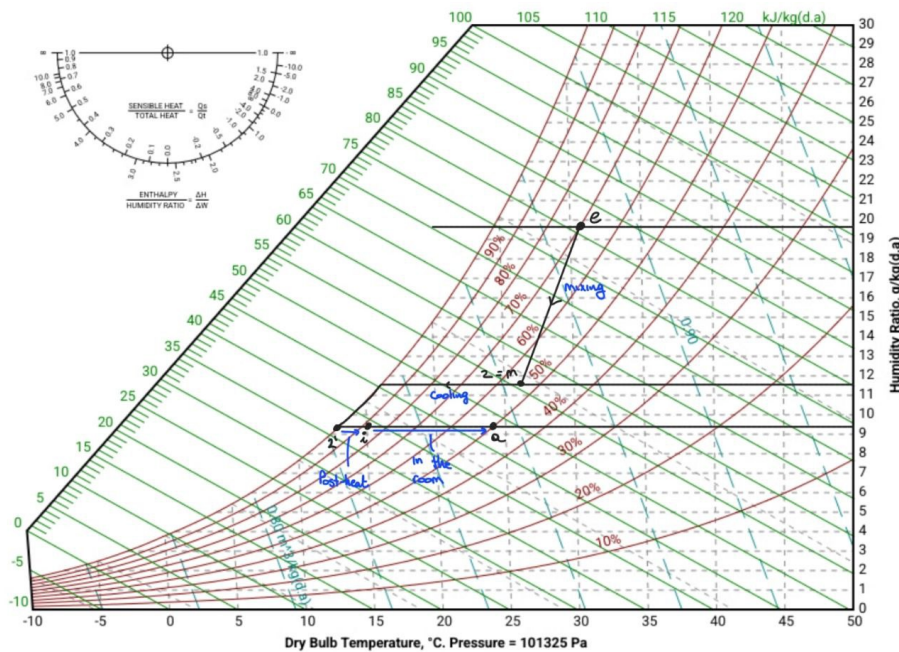


Figure 2.13: HVAC Processes in summer

2.7 Refrigeration Cycle

To ensure comfortable conditions in the department, an air conditioning system is required to maintain appropriate summer and winter temperatures. Therefore, a refrigeration cycle is necessary, which is to be described in the following paragraph. For both summer and winter conditions, some simplifications have been applied:

- Steady-state condition has been assumed, therefore transients have been neglected.
- The condenser and evaporator were not dimensioned, but treated as black boxes.
- The heat exchanger was assumed to operate at constant pressure (isobaric condition).
- The isentropic efficiency of the compressor was assumed to be 0.8.

2.7.1 Summer

The refrigeration system consists of a vapour compression cycle using R134a gas, in accordance with regulations. The fluid properties, including the cycle limits, were evaluated using the COOLPACK software. The cycle analysis was carried out taking into account the simplifications listed above.

The cycle can be described by the following thermodynamic transformation with reference to Figure 2.14:

- 1-2 Compression: the pressure is increased by the work of the compressor, and since the process is not ideal, the entropy also increases.
- 2-3 Condenser: an isobaric process in which heat is released to the surroundings.
- 3-4 Throttling valve: pressure reduction with constant total enthalpy.
- 4-1 Evaporator: an isobaric process in which heat is absorbed from the environment to be cooled.

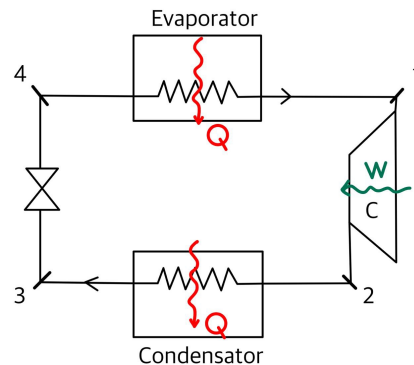


Figure 2.14: Compression cycle schematic

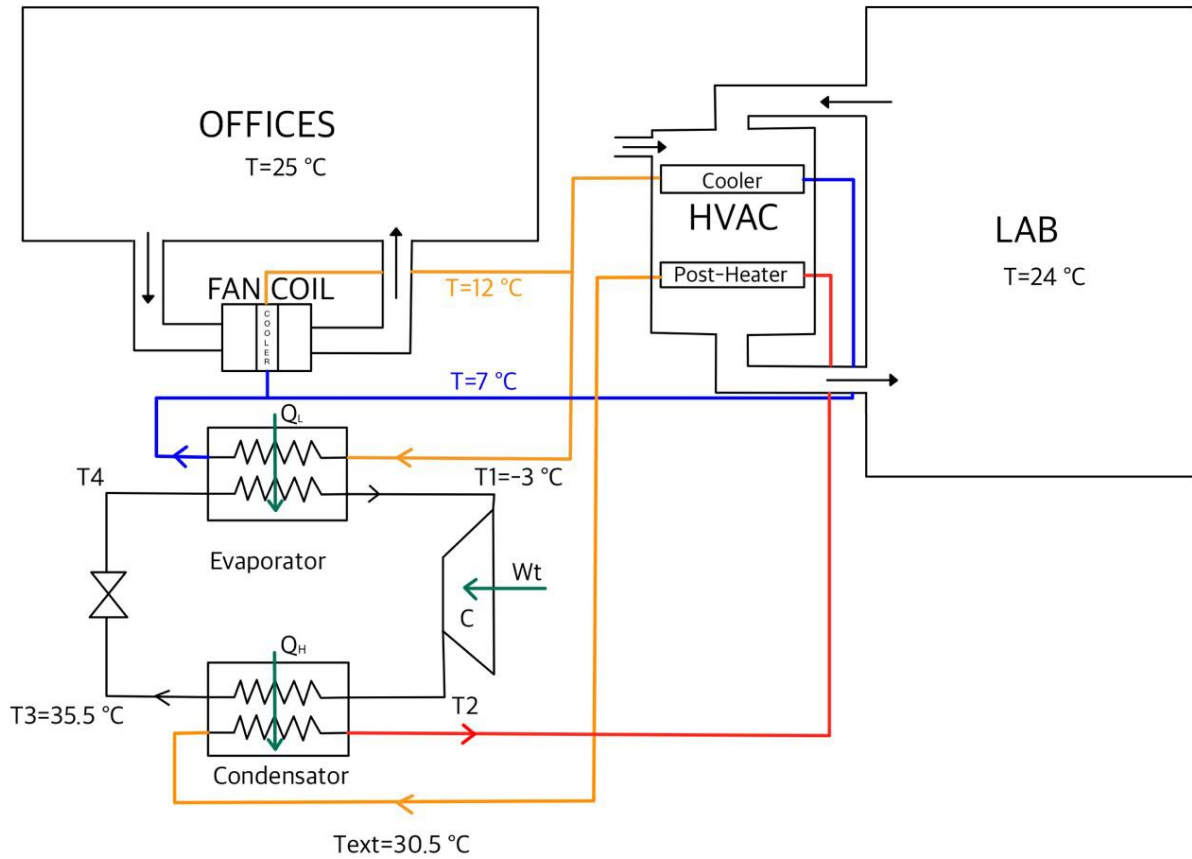


Figure 2.15: Air conditioning system schematic - summer condition

The cycle points were determined according to the following logic:

- **Point 1:** According to fan coil Daikin catalogue [1], the expected inlet water temperature is 7°C. For the R134a vapor temperature, a $\Delta T = -10^{\circ}\text{C}$ lower was chosen; therefore, the temperature at point T1 was set to -3°C.
- **Point 3:** The temperature at point T3 was chosen considering that the external temperature in Turin is estimated to be 30.5°C. A $\Delta T = 5^{\circ}\text{C}$ was applied, resulting in T3 being set to 35.5°C. Moreover, point 3 must lie on the saturated liquid line.
- **Point 4:** In the evaporator, we consider the phase change of the fluid to be isothermal, so the temperature at point T4 will be equal to T1. Considering also that the throttling valve connecting points 3 and 4 is isenthalpic, it is possible to locate point 4.
- **Point 2:** The condenser is considered isobaric, so $p_2 = p_3$. Moreover, the assumed efficiency of 0.8 makes it possible to locate point 2.

Points	1	2	3	4
T [°C]	-3.00	45.50	35.50	-3.00
p [bar]	2.62	9.00	9.00	2.62
h [kJ/kg]	395.41	427.25	249.49	249.49
s [kJ/kgK]	1.72	1.74	1.17	1.18
x	1	gas	0	0.268 (gas+L)

Table 2.18: Refrigeration cycle

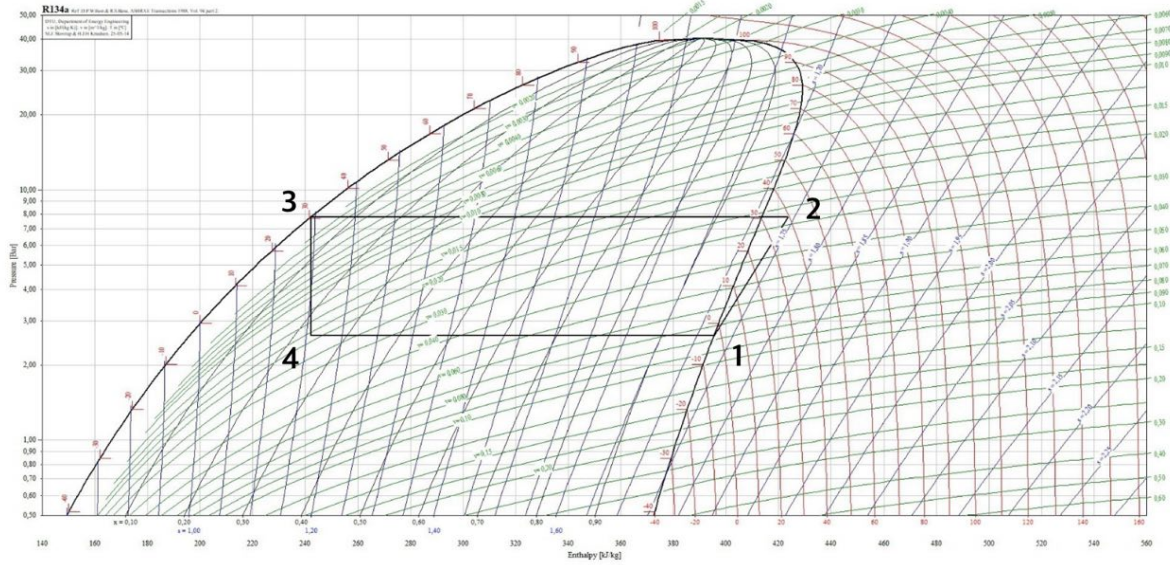


Figure 2.16: Cycle of the summer condition

The adopted convention considers heat as positive, and $\dot{Q}_{2-3}$ is regarded as heat rejected to the environment ($\dot{Q}_H$). The heat input to the evaporator (process 4-1) is the heat extracted from the environment to be cooled ($\dot{Q}_L$), and W_c is the work done by the compressor. The heat to be extracted $\dot{Q}_L$ is considered as the sum of all the contributions from the different areas that make up the department.

$$\dot{Q}_H = \dot{Q}_{2-3}, \quad \dot{Q}_L = \dot{Q}_{4-1}, \quad \dot{W}_c = \text{compressor work rate}$$

Knowing the thermodynamic properties at each point, it is therefore possible to calculate the mass flow rate of the refrigerant vapor:

$$\dot{m}_{\text{R134a}} = \frac{\dot{Q}_L}{h_1 - h_4} = 1.213 \text{ kg/s}$$

Consequently, we can estimate the heat released by the vapor to the environment in the condenser:

$$\dot{Q}_H = \dot{m}_{\text{R134a}} \cdot (h_2 - h_3) = 216 \text{ kW}$$

A portion of this heat can be used to operate the post-heater in the HVAC system.

We can now determine the power requested by the compressor $\dot{W}_C$ as:

$$\dot{W}_C = \dot{Q}_H - \dot{Q}_L = 39 \text{ kW}$$

And finally, the coefficient of performance (COP):

$$COP = \frac{\dot{Q}_L}{\dot{W}_C} = 4.58$$

$\dot{Q}_L[kW]$	177
$\dot{m}_{R134a}[kg/s]$	1.213
$\dot{Q}_H[kW]$	216
$\dot{W}_C[kW]$	39
COP	4.58

Table 2.19: Summary table of main thermodynamic quantities

2.7.2 Winter condition

Under winter conditions, the cycle remains largely similar. Heating is provided by a heater that derives its thermal energy from the heat exchange occurring at the condenser.

The cycle points have been estimated as follows:

- **Point 3:** According to the fan coil catalogue, the inlet water temperature is 65°C. A temperature difference (ΔT) of 10°C has therefore been chosen, which results in a steam temperature at point 3 of $T_3 = 10^\circ\text{C}$. As in the previous case, T_3 is selected on the saturated liquid curve.
- **Point 1:** Considering an external temperature of -8°C, a ΔT of 10°C is maintained. Therefore, the temperature at point 1 is $T_1 = -18^\circ\text{C}$. As in the summer case, it is considered on the saturated vapor line.
- **Point 4:** The expansion valve is assumed to be isenthalpic, so the point is determined through a process at constant enthalpy from point 3 to the temperature T_1 .
- **Point 2:** The condenser is considered isobaric, thus $p_2 = p_3$. Moreover, the assumed efficiency of 0.8 allows the location of point 2.

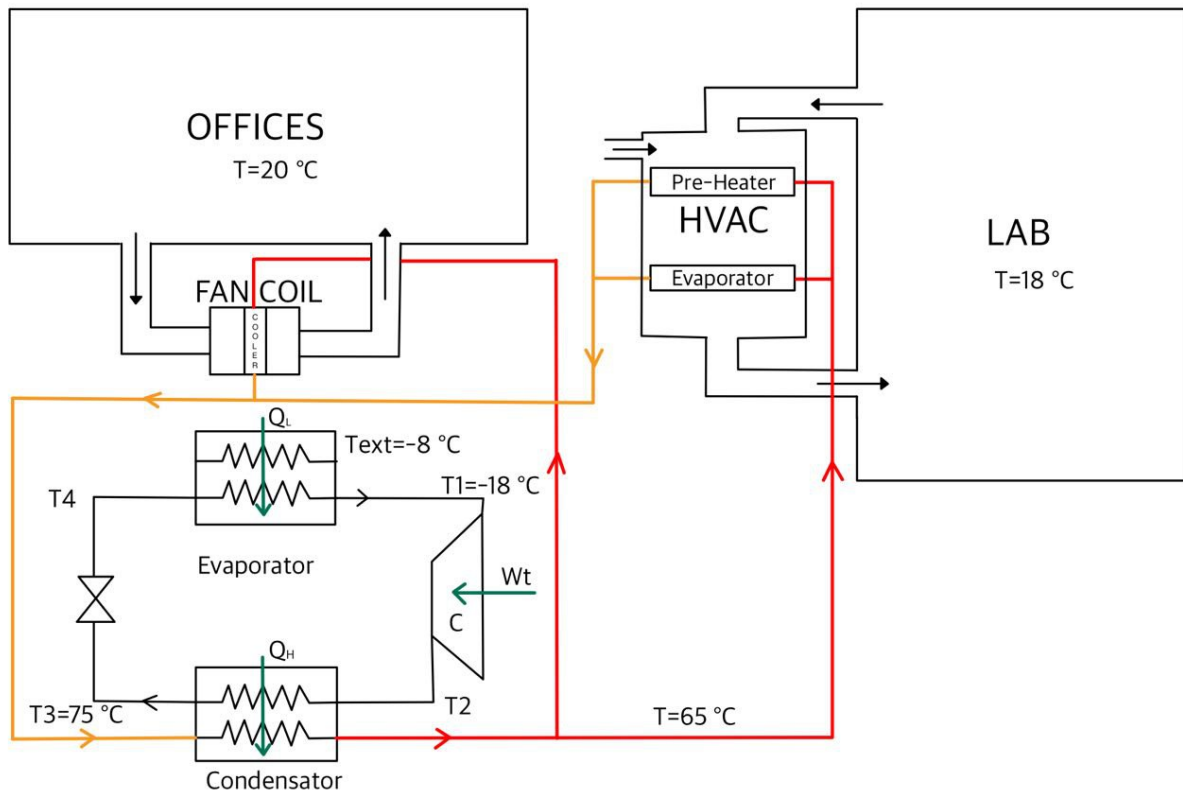


Figure 2.17: Air conditioning system schematic - winter condition

Points	1	2	3	4
T [°C]	-18.00	84.40	75.00	-18.00
p [bar]	1.45	23.62	23.62	1.45
h [kJ/kg]	386.33	444.42	313.27	313.25
s [kJ/kgK]	1.73	1.73	1.36	1.45
x	1	gas	0	0.651 (gas+L)

Table 2.20: Refrigeration cycle

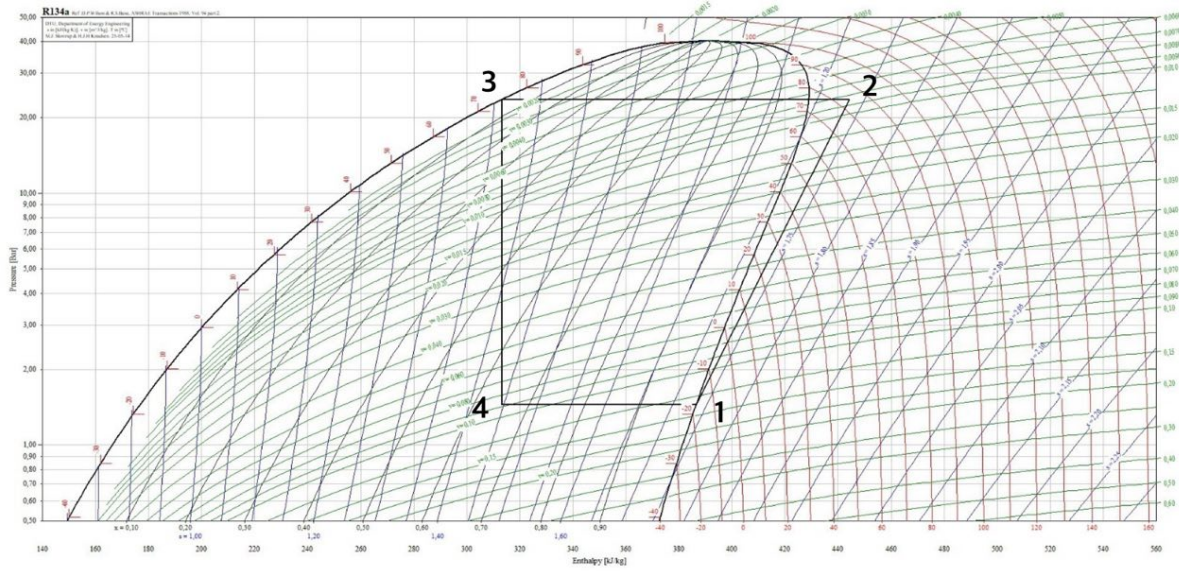


Figure 2.18: Cycle of the winter condition

Once the total thermal load to be supplied to the building environments, $\dot{Q}_H$ (heat released by the gas in the condenser), is estimated, the mass flow rate of R134a vapor can be calculated as:

$$\dot{m}_{\text{R134a}} = \frac{\dot{Q}_H}{h_2 - h_3} = 3.064 \text{ kg/s}$$

It is therefore now possible to calculate the heat exchanged in the evaporator, $\dot{Q}_L$:

$$\dot{Q}_L = \dot{m}_{\text{R134a}}(h_1 - h_4) = 224 \text{ kW}$$

The power supplied to the compressor is:

$$W_c = \dot{Q}_H - \dot{Q}_L = 178 \text{ kW}$$

Consequently, the coefficient of performance (COP) is:

$$COP = \frac{\dot{Q}_H}{W_c} = 2.26$$

$\dot{Q}_H [\text{kW}]$	402
$\dot{m}_{\text{R134a}} [\text{kg/s}]$	3.064
$\dot{Q}_L [\text{kW}]$	224
Compressor work [kW]	178
COP	2.26

Table 2.21: Summary table of main thermodynamic quantities (winter case)

2.8 Pipe sizing

The preliminary dimensioning of the air and water ducts follows the entire rationale suggested by QUADERNI CALEFFI, both with regard to empirical correlations and the value of localized pressure drop coefficients.

Pressure losses are pressure losses (with irreversible transformations of mechanical energy into heat) caused by resistances opposing the motion of a fluid through a conduit. Pressure losses can be continuous or localized:

- continuous losses occur along straight and constant cross section of ducts;
- localized losses occur at special pieces (joints, couplings, bends, branches, confluences, etc.) and components that vary the direction or cross section of fluid flow.

2.8.1 Air

To provide heat-treated air in laboratories, the HVAC is connected to each laboratory by using aluminum pipes, for the sake of simplicity only one type of air duct is considered, precisely tubes have square cross section.

Flow and Velocity

Starting from the mass flow rate calculated in 2.45, we considered the worst-case scenario. and saw that the required mass flow rate had to be much larger, as we saw in 2.29, found by the analysis of winter case HVAC, in order to be more conservative and cover properly the requirements regarding both winter and summer conditions.

The conversion from mass flow rate to volumetric is made according to the following equation: G is:

$$G = \frac{\dot{m}}{\rho} \quad \left[\frac{\text{m}^3}{\text{s}} \right]$$

Through this conversion we could also evaluate the different speeds of the fluid into the pipes, by the usage of this formula:

$$u = \frac{G}{A}$$

Where:

- u = fluid velocity $[m/s]$
- G = volumetric flow rate $[m^3/s]$
- A = cross-sectional area $[m^2]$

To meet comfort requirements and ensure uniform distribution throughout the system, a target air velocity of $1.5m/s$ was imposed.

Air Density

The density has been modified, considering the temperature effect and a preliminary modify of barometric pressure, considering the height of Turin from the sea level.

$$\rho = 1.293 \cdot \left(\frac{P_b}{1.01325} \right) \cdot \frac{273}{273 + t} = 1.29 \text{ kg m}^{-3}$$

$$\text{where } P_b = -0.1125 \cdot H + 1011.5 = 0.984 \text{ Pa} \quad , \quad H = 239 \text{ m} \quad \text{and} \quad t = -8^\circ \text{C}$$

Kinematic Viscosity

Due to the behavior of the piping system, that requires a fluid in motion, the focus must be also on the study of viscosity. The kinematic viscosity is a fundamental parameter if an analysis of head losses, and is obtained by the dynamic viscosity and the density of air, that has been calculated before. We calculated the new absolute kinematic viscosity by using the formula found in "Quaderni Caleffi" reported below:

$$\nu = \left(\frac{1.53}{\rho} \right) \cdot 10^{-6} \cdot \frac{(273 + t)^{1.5}}{413 + t} = 1.30 \text{e-}5 \text{ m}^2 \text{ s}^{-1}$$

where:

- ν = kinematic viscosity of air [$\text{m}^2 \text{s}^{-1}$]
- t = temperature= -8°C
- ρ = air density = 1.33 kg m^{-3}

Roughness

It is a quantity used to account for the average irregularities of a surface. It is generally denoted by the symbol ε . For commercial ducts that convey air, very smooth ones are adopted, thus $\varepsilon = 0.03 \text{ mm}$

Reynolds Number

It is a nondimensional parameter, useful to determine how the fluid moves inside our ducts. We can evaluate it through this formula:

$$Re = \frac{u \cdot D}{\nu}$$

Where:

- Re = dimensionless Reynolds number;

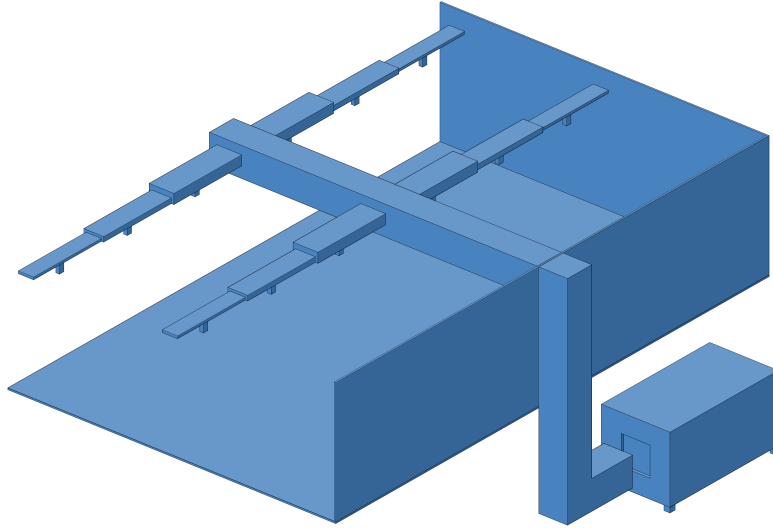


Figure 2.19: Air piping scheme

- v = velocity of the fluid within the channel, m/s ;
- D = characteristic length, m ;
- ν = kinematic viscosity of the air, m^2/s ;

In particular, three different domains are identified:

- laminar: $Re < 2000$
- Transition $2000 < Re < 2500$
- Turbulent: $Re > 2500$

The value for the Reynold's number is shown in the table below 2.23

Network Layout

In this study, a double-T layout was adopted for the hydraulic distribution network, with the main branch positioned centrally to ensure balanced flow distribution.

Equivalent Diameter Calculation

Starting from the imposed velocity and the required flow rate for each section, the internal diameters of the main and secondary branches were calculated. First of all, since the speed of fluid is imposed, the sections for each duct are obtained by:

$$A = \frac{G}{v}$$

Then, the process required the evaluation of the equivalent diameter: Since the task imposed to assume square sections, the conversion of the square ducts to equivalent circular section is provided by the Huebscher formula:

$$D_{\text{eq}} = 1.3 \cdot \frac{(a \cdot b)^{0.625}}{(a + b)^{0.25}}$$

Where:

- D_{eq} is the equivalent diameter.
- a and b are the lengths of the sides of a rectangular duct.

Case	a [m]	b [m]	D_{eq} [m]
1	1.46	2.2	1.95
2	1.04	1.56	1.39
3	0.73	1.1	0.98
4	0.613	0.92	0.81
5	0.42	0.63	0.56

Table 2.22: Dimensions and equivalent diameters (Hübscher Formula)

These obtained diameters were then rounded to the nearest standard commercial values to ensure compatibility with conventional piping. Once the final diameters were set, the actual velocities in the various branches were re-evaluated based on the modified geometry.

Section	G [m ³ s ⁻¹]	V_{imp} [m s ⁻¹]	D_{req} [m]	D_{tab} [m]	V_{eff} [m s ⁻¹]	Re
1	4.55	1.50	1.95	2	1.45	2.24e5
2	2.28	1.50	1.39	1.4	1.48	1.60e5
3	1.13	1.50	0.98	1	1.45	1.12e4
4	0.76	1.50	0.81	0.9	1.19	8.29e4
5	0.38	1.50	0.56	0.6	1.34	6.21e4

Table 2.23: Flow parameters and diameter calculations

Friction Factor Calculation

The dimensionless friction factor was then estimated using the empirical formula provided by Caleffi, consistent with typical design practices for turbulent flow in smooth pipes, since the roughness chosen is equal to $\varepsilon = 0.03\text{mm}$.

$$Fa^* = 0.11 \cdot \left(\frac{\varepsilon}{D} + 192.3 \cdot \frac{D \cdot \nu}{G} \right)^{0.25}$$

$$Fa = \begin{cases} Fa^* & \text{if } Fa^* \geq 0.018 \\ 0.85 \cdot Fa^* + 0.0028 & \text{otherwise} \end{cases}$$

Where:

- Fa^* = conventional friction factor (dimensionless)
- Fa = friction factor (dimensionless)
- ε = roughness [mm]
- ν = air kinematic viscosity [$\text{m}^2 \text{s}^{-1}$]
- G = flow rate [$\text{m}^3 \text{h}^{-1}$]
- D = internal diameter [mm]

The continuous pressure drop per unit length is given by:

$$r = 6.254 \cdot 10^7 \cdot Fa \cdot \rho \cdot \frac{G^2}{D^5}$$

Where:

- r = continuous pressure drop per unit length [Pa m^{-1}]
- Fa = friction factor (dimensionless)
- ρ = density [kg m^{-3}]
- G = flow rate [$\text{m}^3 \text{h}^{-1}$]
- D = internal diameter [mm]

Section	Fa^*	Fa	r [Pa m^{-1}]
1	1.50e−2	1.56e−2	1.03e−2
2	1.64e−2	1.68e−2	1.64e−2
3	1.81e−2	1.81e−2	2.37e−2
4	1.90e−2	1.90e−2	1.90e−2
5	2.00e−2	2.00e−2	3.88e−2

Table 2.24: Friction factors and pressure drop values

Continuous (distributed) pressure losses were computed for each segment using the Darcy-Weisbach equation, taking into account the equivalent friction factor and the length of each duct section.

Where:

Section	r [Pa m ⁻¹]	L [m]	N	ΔP [Pa]
1	1.03e-2	14.25	1	0.147
2	1.64e-2	16.50	1	0.27
3	2.37e-2	4.38	4	0.416
4	1.90e-2	4.38	4	0.33
5	3.88e-2	4.38	4	0.68

Table 2.25: Pressure drop calculations by section

- r = pressure drop per unit length
- L = length of the section
- N = number of parallel elements
- ΔP = section pressure drop

Thus having a total pressure loss:

$$\Delta P_{tot} = 1.85 \text{ Pa} \quad (2.53)$$

Localized Losses

In addition, localized pressure losses were calculated for each branch.

$$z = \xi \cdot \rho \cdot \frac{v^2}{2}$$

Where:

- z = **localized pressure loss** [Pa]
- ξ = localized loss coefficient (dimensionless)
- ρ = density [kg m⁻³]
- v = velocity [m s⁻¹]

Table 2.26: Localized Losses by Section

Type	Section	ξ	N	ΔP [Pa]
2	1	0.2	3	0.815
3	1	0.7	2	1.902
1	1	0.2	2	0.566
2	2	1.4	2	3.961
3	2	1.4	1	1.980
2	3	0.2	4	1.087
4	3	0.8	4	4.347
2	4	0.2	4	0.736
4	4	0.8	4	2.945
4	5	0.8	4	3.727
5	3	4.4	4	20.459
6	4	1.8	4	8.386
7	5	0.6	4	2.795

The localized pressure drop equals:

$$\Delta P_{\text{conc}} = 53.75 \text{ Pa}$$

The total pressure drop for the system was obtained by summing all distributed, localized and global losses.

The total pressure drop in 1 circuit equals:

$$\Delta P_{\text{circ}} = 55.59 \text{ Pa}$$

Since we have two storeys, we can multiply times 2 the ΔP_{circ} , so that we obtain

$$\Delta P_{\text{global}} = 111.18 \text{ Pa}$$

So, the power is

$$P = G \cdot \Delta P_{\text{glob}} = 1012.36 \text{ W}$$

Fitting Type Codes

The total pressure drop for the system was obtained by summing all distributed, localized and global losses. Since the analysis was performed for a single floor, the final values were doubled to account for the complete system configuration involving both floors. Then, the final Power needed is the product of the Induced Volumetric flow rate times the deltaP to be satisfied, to provide the necessary flow rate of air in the two laboratories, with the right pressure at each outlet.

Fitting Type Codes	
Where:	
	Restriction = 2
	90° curve = 1
	T-junction = 3
	outlet = 4
	Centrally-mounted disc throttle valve (30°) = 5
	Centrally-mounted disc throttle valve (20°) = 6
	Centrally-mounted disc throttle valve (10°) = 7

2.8.2 Water

The most conservative and worst-case scenario of supplying water to the 5th floor is considered, due to the higher thermal load, equal to 68 006 W. In order to provide the laboratories with the appropriate water flow rate, a properly designed piping system is required. The key quantities (related to both the fluid and the conduits) involved in the head loss calculations are:

- velocity, v ,
- density, ρ ,
- kinematic viscosity, ν ,
- roughness, ϵ ,
- Reynolds number, Re

These parameters will be analyzed in the following subsections.

To properly design the water piping system, the worst-case thermal load condition—considering both winter and summer scenarios—was evaluated across all five floors of the building. Based on a total thermal load of 68 006 W and a total of 19 offices to be served, the required mass flow rate of water was calculated using:

$$\dot{G} = \frac{\Phi_{\text{tot}}}{c_{\text{water}} \cdot \Delta t} = 0.57 \text{ kg s}^{-1} \quad (2.54)$$

Where:

- $\dot{G}$ = mass flow rate [kg s^{-1}]
- Φ_{tot} = total heat flow rate [W]
- c_{water} = specific heat capacity of water = $4186 \text{ J kg}^{-1} \text{ K}^{-1}$
- Δt = temperature difference = 28 K

Water Density

Since a conversion from mass flow rate to volumetric flow rate is necessary, the density of water is calculated, taking into account the effect of temperature, using the empirical formula provided in 'Quaderni Caleffi':

$$\rho = 1000.18576 + 0.007136 \cdot t - 0.005718 \cdot t^2 + 0.0001468 \cdot t^3 \quad (2.55)$$

For $t = -8^\circ\text{C}$:

$$\rho = 999.75 \text{ kg m}^{-3}$$

This allows us to convert the mass flow rate into volumetric flow rate:

$$Q = \frac{\dot{G}}{\rho} = \frac{0.573}{999.76} = 0.000573 \text{ m}^3 \text{ s}^{-1}$$

Flow and Velocity

Using the previously computed volumetric flow rate, the fluid velocity can be calculated with:

$$u = \frac{Q}{A}$$

Where:

- u = fluid velocity [m s^{-1}]
- Q = volumetric flow rate [$\text{m}^3 \text{ s}^{-1}$]
- A = cross-sectional area of the pipe [m^2]

To ensure comfort and efficient distribution throughout the system, a target design velocity of 1.5 m s^{-1} was imposed. This allows calculation of the required pipe diameter.

Kinematic Viscosity

Since the system involves fluid motion, viscosity plays a crucial role in head loss calculations. The kinematic viscosity of water is defined as:

$$\nu = \frac{\mu}{\rho}$$

For engineering calculations, the following empirical formula for kinematic viscosity at a given temperature t in $^\circ\text{C}$ is used:

$$\nu = (1.67952 - 0.042328 \cdot t + 0.000499 \cdot t^2 - 0.00000214 \cdot t^3) \times 10^{-6} \quad (2.56)$$

Substituting $t = -8^\circ\text{C}$:

$$\nu = 2.05\text{e-}6 \text{ m}^2 \text{ s}^{-1}$$

Roughness

A pipe roughness of $\epsilon = 0.003 \text{ mm}$ is assumed, typical for commercial steel or plastic piping used in HVAC systems.

Reynolds Number

The Reynolds number is computed using:

$$Re = \frac{u \cdot D}{\nu}$$

Where:

- Re = Reynolds number [dimensionless]
- u = fluid velocity [m s^{-1}]
- D = internal diameter of the pipe [m]
- ν = kinematic viscosity [$\text{m}^2 \text{ s}^{-1}$]

This will determine the flow regime (laminar or turbulent), which is crucial for further calculations of head losses. As in the previous case, the values for the Reynold's numbers are found in the table 2.27

Network Layout

Flow rates in each section of the circuit were estimated using a double-T layout, similar to the one adopted in the air distribution system. In this configuration, the final segment was designed to serve three offices instead of two. Consequently, each successive branch carries a flow rate reduced by $\frac{2}{19}$ of the total, corresponding to the demand of two offices.

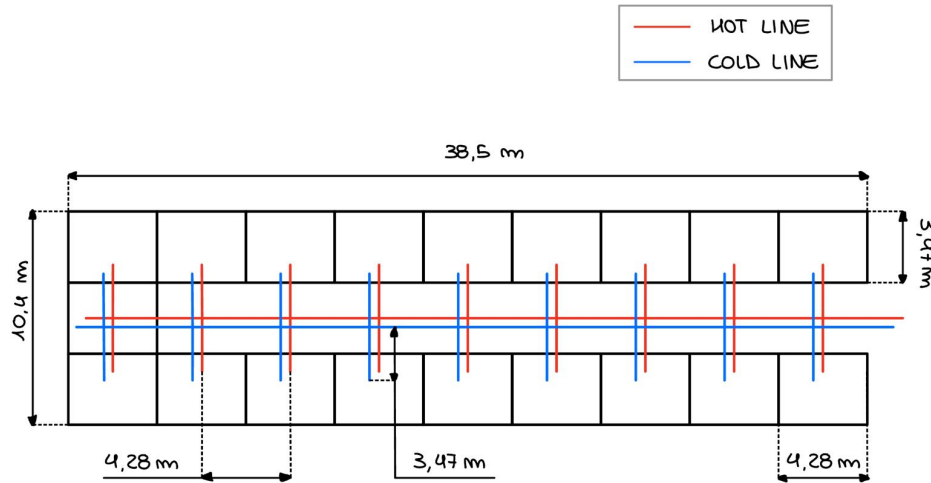


Figure 2.20: Scheme of water pipes

By fixing the water velocity and evaluating the required mass flow rates, the necessary pipe diameters for each segment were calculated. These computed diameters were then rounded to the nearest standard commercial sizes to ensure compatibility with available piping products.

Following the diameter adjustment, actual velocities in the modified sections were re-evaluated to reflect the adopted commercial diameters.

Table 2.27: Flow and Reynolds Number Analysis

Row	G [m ³ /s]	V _{required} [m/s]	D _{required} [m]	D _{tab} [m]	V [m/s]	Re
1	5.73E-04	1.50	0.022	0.03	0.811	1.19E+04
2	5.13E-04	1.50	0.02087	0.03	0.726	1.06E+04
3	4.53E-04	1.50	0.0196	0.02	1.44	1.41E+04
4	3.92E-04	1.50	0.01825	0.02	1.24	1.22E+04
5	3.32E-04	1.50	0.01679	0.02	1.06	1.03E+04
6	2.72E-04	1.50	0.01518	0.02	0.86	8.43E+03
7	2.11E-04	1.50	0.01339	0.02	0.67	6.56E+03
8	1.51E-04	1.50	0.00877	0.02	0.48	4.68E+03
9	9.05E-05	1.50	0.00506	0.01	1.15	5.62E+03
10	3.02E-05	1.50	0.00848	0.01	0.38	1.87E+03

Pressure Drop Calculation

The pressure drop per unit length r was calculated using the empirical relation applicable to smooth pipes:

$$r = 0.241 \cdot \nu^{0.25} \cdot \rho \cdot \frac{G^{1.75}}{D^{4.75}} \quad (2.57)$$

where:

- r : pressure drop per unit length [Pa m^{-1}]
- ν : kinematic viscosity [$\text{m}^2 \text{s}^{-1}$]
- ρ : fluid density [kg m^{-3}]
- G : volumetric flow rate [$\text{m}^3 \text{s}^{-1}$]
- D : pipe diameter [m]

Distributed pressure losses for each section were then determined by multiplying the pressure drop per unit length by the length of the corresponding duct segment.

Table 2.28: Pressure Drop Calculations by Section

Section	Re	L [m]	r [Pa/m]	N	ΔP [Pa]
1	1.19E+04	4.282	331.85	1	1.42E+03
2	1.06E+04	4.282	273.15	1	1.50E+03
3	1.41E+04	4.282	1505.6	1	6.45E+03
4	1.22E+04	4.282	1172	1	5.02E+03
5	1.03E+04	4.282	874.96	1	3.75E+03
6	8.43E+03	4.282	615.85	1	2.64E+03
7	6.56E+03	4.282	396.71	1	1.70E+03
8	4.68E+03	4.282	220.16	1	9.43E+02
9	5.62E+03	4.282	2423.29	1	1.04E+04
10	1.87+03	4.282	354.36	19	2.33+04

The total distributed pressure loss is therefore:

$$\Delta P_{\text{tot}} = 56.8 \text{ kPa}$$

Localized Losses

Localized pressure losses were computed using the standard formula:

$$z = \xi \cdot \rho \cdot \frac{v^2}{2}$$

where:

- z : localized pressure loss [Pa]
- ξ : loss coefficient (dimensionless)
- ρ : fluid density [kg m^{-3}]
- v : velocity [m s^{-1}]

Table 2.29: Pressure Loss Calculations

Type	Section	ζ	N	ΔP [Pa]
1	1	0.5	1	164.49
2	1	1	2	657.96
4	1	0.4	2	263.18
1	2	0.5	1	131.68
2	2	1	2	526.73
1	3	0.5	1	519.02
2	3	1	2	2076.067
1	4	0.5	1	389.84
2	4	1	2	1559.36
2	5	0.5	1	279.12
2	5	1	2	1116.46
1	6	0.5	1	186.85
2	6	1	2	747.38
1	7	0.5	1	113.03
2	7	1	2	452.12
1	8	0.5	1	57.67
2	8	1	2	230.67
1	9	0.5	1	332.17
2	9	1	2	1328.68
3	10	4	19	5609.99
5	1.2	1.5	1	888.53
5	3.4	2	1	3635.4
5	5.6	2	1	1863.85
5	7.8	2	1	682.8
5	9.10	3.5	1	1476.3

The total localized pressure loss is:

$$\Delta P_{\text{loc}} = 2.5 \times 10^4 \text{ Pa}$$

Total Pressure Drop and Required Power

The global pressure drop, accounting for both distributed and localized losses, is:

$$\Delta P_{\text{global}} = 1.64 \times 10^5 \text{ Pa}$$

As the system includes both flow and return circuits, this value was doubled.

Additionally, the geodetic pressure difference from the ground to the fifth floor was included:

$$\Delta P_{\text{height}} = \rho \cdot g \cdot H = 1.47 \times 10^5 \text{ Pa}$$

The total system pressure drop becomes:

$$\Delta P_{\text{tot}} = \Delta P_{\text{height}} + \Delta P_{\text{global}} = 3.11 \times 10^5 \text{ Pa}$$

Finally, the power required to circulate the water is given by:

$$P = G \cdot \Delta P_{\text{tot}} = 178.21 \text{ W}$$

Classification of Localized Pressure Loss Elements

The following table lists the types of elements used in the localized loss analysis:

Table 2.30: Localized Loss Element Classification

Type ID	Element	Type ID	Element
1	Restriction	2	Double Branch
3	Angle-type Radiator Valve	4	90° Bend
5	butterfly valve		

Chapter 3

Energy Analysis

3.1 Cost analysis

3.1.1 Summer Case

The Coefficient of Performance (COP) of the refrigeration cycle can be obtained from the previous calculations:

$$COP = \frac{\dot{Q}_H}{\dot{W}_c} \quad (3.1)$$

where:

- $\dot{Q}_H = 216 \text{ kW}$ is the thermal power removed from the offices and lab (cooling capacity)
- $\dot{W}_c = 39 \text{ kW}$ is the power requested by the compressor

$$COP = \frac{216}{39} = 4.58 \quad (3.2)$$

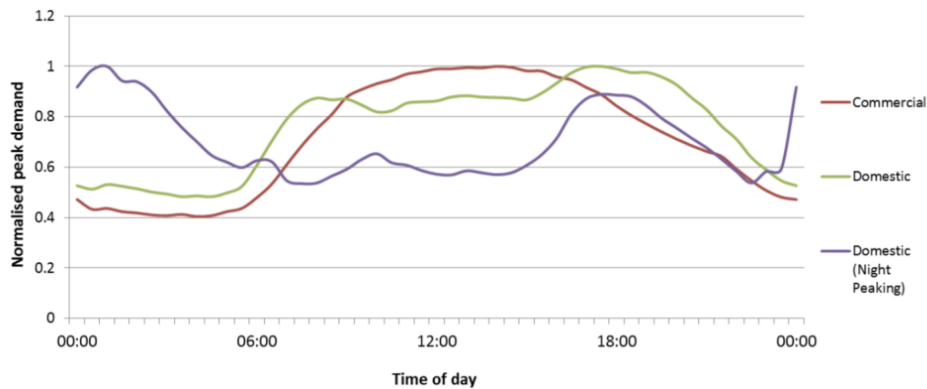


Figure 3.1: Normalized load conditions (CITE)

In order to perform a cost analysis, it is necessary to evaluate the average consumption of the building. To do this, we consider the curve represented in Figure 3.1, and consider

the “Domestic” case. Using <https://automeris.io/wpd/> we are able to extract accurately the normalized peak demand at every hour; this is used for the computation of the energy used in a single day.

By recalling that for the summer, the maximum compressor work is $W_c = 39 \text{ kWh}$ we can summarize for every hour the energy consumption:

Energy SUMMER (kWh)		
HOURL	Domestic	SUMMER
00:00	52%	20.1 kWh
01:00	52%	20.1 kWh
02:00	52%	20.1 kWh
03:00	49%	18.9 kWh
04:00	48%	18.5 kWh
05:00	49%	18.9 kWh
06:00	56%	21.6 kWh
07:00	75%	29.0 kWh
08:00	87%	33.6 kWh
09:00	87%	33.6 kWh
10:00	83%	32.1 kWh
11:00	84%	32.4 kWh
12:00	86%	33.2 kWh
13:00	88%	34.0 kWh
14:00	87%	33.6 kWh
15:00	87%	33.6 kWh
16:00	92%	35.5 kWh
17:00	99%	38.2 kWh
18:00	100%	38.6 kWh
19:00	98%	37.8 kWh
20:00	94%	36.3 kWh
21:00	85%	32.8 kWh
22:00	74%	28.6 kWh
23:00	61%	23.6 kWh
TOTAL		704.8 kWh

Table 3.1: Hourly energy consumption - Summer

Next, we can compute the daily cost of energy with the following equation:

$$TC_d = E_d \cdot EC = 704.8 \text{ kWh} \cdot 0.1232 \text{ €/kWh} \quad (3.3)$$

where:

- TC_d is the total cost of electrical energy in one day [*eur/day*]
- E_d is the energy requested per day by the compressor [*kWh*]
- EC is the energy cost [*€/kWh*] from *OctopusEnergy*:
<https://octopusenergy.it/le-nostre-tariffe>

3.1.2 Winter Case

With a similar approach for the winter case, the Coefficient of Performance (COP) of the refrigeration cycle can be obtained from the previous calculations:

$$COP = \frac{\dot{Q}_H}{\dot{W}_c} \quad (3.4)$$

where:

- $\dot{Q}_H = 402 \text{ kW}$ is the thermal power released to the offices and lab (heating capacity)
- $\dot{W}_c = 178 \text{ kW}$ is the power requested by the compressor

$$COP = \frac{402}{178} = 2.26 \quad (3.5)$$

By recalling that for winter the maximum compressor work is $W_c = 178 \text{ kWh}$, the hourly energy consumption is reported in Table 3.2.

Analogously to the summer case, we can compute the daily cost of energy with the following equation:

$$TC_d = E_d \cdot EC = 3247 \text{ kWh} \cdot 0.1232 \text{ €/kWh} \quad (3.6)$$

Energy WINTER (kWh)		
HOURL	Domestic	WINTER
00:00	52%	92.5 kWh
01:00	52%	92.5 kWh
02:00	52%	92.5 kWh
03:00	49%	87.2 kWh
04:00	48%	85.4 kWh
05:00	49%	87.2 kWh
06:00	56%	99.6 kWh
07:00	75%	133.4 kWh
08:00	87%	154.8 kWh
09:00	87%	154.8 kWh
10:00	83%	147.7 kWh
11:00	84%	149.4 kWh
12:00	86%	153.0 kWh
13:00	88%	156.6 kWh
14:00	87%	154.8 kWh
15:00	87%	154.8 kWh
16:00	92%	163.7 kWh
17:00	99%	176.1 kWh
18:00	100%	177.9 kWh
19:00	98%	174.4 kWh
20:00	94%	167.2 kWh
21:00	85%	151.2 kWh
22:00	74%	131.7 kWh
23:00	61%	108.5 kWh
TOTAL		3247.0 kWh

Table 3.2: Hourly energy consumption - Winter

3.2 Efficiency improvements

Let us first recall the **total building** thermal loads computed in Part 0:

Summer Thermal Load [kW]	Winter Thermal Load [kW]
177	402

Table 3.3: Total building thermal loads before changes

These are quite high, and we want to reduce them. To reduce the thermal load, we have decided to act on the opaque surfaces by increasing the isolation present. This will lower the overall transmissivity. To this end, we have decided to use an expanded polystyrene (EPS) panel of 8 cm. This material has a very low thermal conductivity $\lambda = 0.035 \frac{W}{(m \cdot K)}$ at room temperature [2].

Let us make the following 1D *electrical resistance analogy*:

where we can recall that the *electrical resistance* is the inverse of the transmissivity:

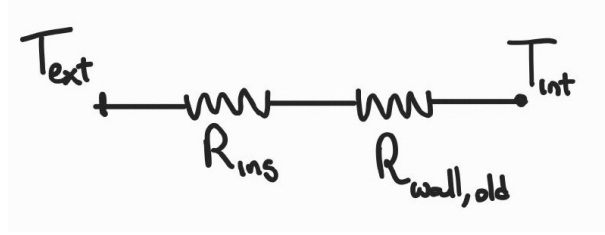


Figure 3.2: 1D electrical resistance analogy

$$R = \frac{1}{U} \left[\frac{m^2 K}{W} \right]$$

The resistance of the new insulating panels is:

$$\begin{aligned} R_{ins} &= \frac{s}{\lambda} \\ &= \frac{0.08}{0.035} \\ &= 2.29 \frac{m^2 K}{W} \end{aligned} \tag{3.7}$$

We can compute the old wall thermal resistance:

$$\begin{aligned} R_{wall,old} &= \frac{1}{U_{wall,old}} \\ &= \frac{1}{1.41} \\ &= 0.71 \frac{m^2 K}{W} \end{aligned}$$

Next we can consider the new wall resistance as the sum of these resistances:

$$R_{wall,new} = R_{wall,old} + R_{ins} = 2.99 \frac{m^2 K}{W} \tag{3.8}$$

And finally, we can compute the new wall transmissivity:

$$U_{wall,new} = \frac{1}{2.99} = 0.334 \frac{W}{m^2 K} \tag{3.9}$$

This is a remarkable decrease from the previous value, it could further be reduced by considering a thicker insulating panel, for example 10 or even 12 cm, but we have decided to move forward with a 8 cm panel.

By performing an analogous procedure, we can consider to insulate the roof with the same material and thickness, resulting in a new roof transmissivity:

$$U_{roof,new} = 0.327 \frac{m^2 K}{W} \tag{3.10}$$

With these new values, the same computations performed in Part 0 were performed in order to compute the new thermal loads:

Summer:

Office Thermal Load [kW]	79
Laboratory Thermal Load [kW]	76
Total Building Thermal Load [kW]	155
Improvement from before	13%

Table 3.4: Total building summer thermal loads after changes

Winter:

Office Thermal Load [kW]	−191
Laboratory Thermal Load [kW]	−109
Total Building Thermal Load [kW]	−300
Improvement from before	25%

Table 3.5: Total building winter thermal loads after changes

3.2.1 Cost analysis - Summer

By considering the same heat load curve as before, and considering the same previously defined points of the thermodynamic cycles, thus changing only the mass flows, the same procedure as before was done for the cost analysis, both in summer and in winter. In summer the new cycle is characterized by the following values:

$\dot{Q}_L[kW]$	155
$\dot{m}_{R134a}[kg/s]$	1.061
$\dot{Q}_H[kW]$	189
$\dot{W}_C[kW]$	34
COP	4.58

Table 3.6: Summary table of new main thermodynamic quantities in summer

From here, we can compute the new daily consumption:

By considering the same cost of electricity we get a daily cost of:

$$\begin{aligned} \text{Cost of Electricity} &= 616.5 \cdot 0.1232 \\ &= 75.95 \text{ €/day} \end{aligned} \tag{3.11}$$

We can note the following two things:

- Monthly cost: 2354.38 €
- Money saved every month: 337 €

Energy SUMMER		
HOURL	Domestic	Compressor Work
00:00	52 %	17.6 kWh
01:00	52 %	17.6 kWh
02:00	52 %	17.6 kWh
03:00	49 %	16.6 kWh
04:00	48 %	16.2 kWh
05:00	49 %	16.6 kWh
06:00	56 %	18.9 kWh
07:00	75 %	25.3 kWh
08:00	87 %	29.4 kWh
09:00	87 %	29.4 kWh
10:00	83 %	28.0 kWh
11:00	84 %	28.4 kWh
12:00	86 %	29.0 kWh
13:00	88 %	29.7 kWh
14:00	87 %	29.4 kWh
15:00	87 %	29.4 kWh
16:00	92 %	31.1 kWh
17:00	99 %	33.4 kWh
18:00	100 %	33.8 kWh
19:00	98 %	33.1 kWh
20:00	94 %	31.8 kWh
21:00	85 %	28.7 kWh
22:00	74 %	25.0 kWh
23:00	61 %	20.6 kWh
TOTAL		616.5 kWh

3.2.2 Cost analysis - Winter

In winter the procedure is analogous:

$\dot{Q}_H[kW]$	300
$\dot{m}_{R134a}[kg/s]$	2.288
$\dot{Q}_L[kW]$	167
$\dot{W}_C[kW]$	133
COP	2.26

Table 3.7: Summary table of new main thermodynamic quantities in winter

Energy WINTER		
HOURL	Domestic	Heat Pump
00:00	52 %	69.1 kWh
01:00	52 %	69.1 kWh
02:00	52 %	69.1 kWh
03:00	49 %	65.1 kWh
04:00	48 %	63.8 kWh
05:00	49 %	65.1 kWh
06:00	56 %	74.4 kWh
07:00	75 %	99.6 kWh
08:00	87 %	115.6 kWh
09:00	87 %	115.6 kWh
10:00	83 %	110.3 kWh
11:00	84 %	111.6 kWh
12:00	86 %	114.3 kWh
13:00	88 %	116.9 kWh
14:00	87 %	115.6 kWh
15:00	87 %	115.6 kWh
16:00	92 %	122.2 kWh
17:00	99 %	131.5 kWh
18:00	100 %	132.9 kWh
19:00	98 %	130.2 kWh
20:00	94 %	124.9 kWh
21:00	85 %	112.9 kWh
22:00	74 %	98.3 kWh
23:00	61 %	81.0 kWh
TOTAL		2424.7 kWh

$$\begin{aligned}
 \text{Cost of Electricity} &= 2424.7 \cdot 0.1232 \\
 &= 298.72 \text{ €/day}
 \end{aligned}
 \tag{3.12}$$

We can note the following two things:

- Monthly cost: 9260.27 €
- Money saved every month: 3140.57 €

Chapter 4

Exergy Analysis

The exergy analysis is useful method to identify the components that contribute most significantly to irreversibilities. This diagnostic phase provides the foundation for the subsequent optimization procedures and, if necessary, for the redesign of critical sections of the studied system.

The exergy analysis implies the definition of specific boundaries of the system or the component under evaluation. As with energy analysis, it is essential to first identify a control volume.

The exergy equation at steady state for a control volume can be written as below. It is reminded that it does not give any additional information with respect to the first and second principles of thermodynamics because it is just a linear combination of the two of them.

$$\sum_k \phi_k \cdot \left(1 - \frac{T_0}{T_k}\right) - W_t = \sum_i^{\text{outlet doors}} G_i \cdot b_i - \sum_j^{\text{inlet doors}} G_j \cdot b_j + T_0 \cdot \psi_{irr}$$

A reference ‘dead-state’ (denoted with the subscript ‘0’) corresponds to the external environment has to be defined in both summer and winter working conditions.

4.1 Summer case

Definition of dead state conditions

To perform a correct exergy analysis, the first thing to do is to define the dead state conditions. This choice is arbitrary and in this case the dead state is defined as the external conditions

$$T_{\text{dead,state}} = 30.5 \text{ }^{\circ}\text{C} \quad P_{\text{dead,state}} = 101324 \text{ Pa}$$

To simplify the notation, the dead state will be indicated with the condition 0. For humid air conditions, one parameter that will be used is

$$\bar{x}_0 = \frac{M_a}{M_v} \cdot x_0 = \frac{28.97}{18.015} \cdot x_0$$

x_0 was previously defined:

$$x_0 = 0.02 \frac{\text{kg}_{\text{water}}}{\text{kg}_{\text{dry_air}}}$$

For what concerns the refrigerating fluid R134a, enthalpy and entropy in dead state condition are defined using Coolpack:

$$h_0 = 429.14 \frac{\text{kJ}}{\text{kg}} \quad s_0 = 1.92 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

In conclusion, in order to evaluate the exergy of water, the incompressible fluid assumption will be made and so the only information needed are the temperature and the pressure of dead state.

Dead state water	
T_0 (°C)	30.5
P_0 (Pa)	101325

Dead state humid hair	
T_0 (°C)	30.5
P_0 (Pa)	101325
x_0*	0.031550326

Dead state R134a	
T_0 (°C)	30.5
P_0 (Pa)	101325
Enthalpy	429.14
Entropy	1.9155

4.1.1 HVAC System

Cooler

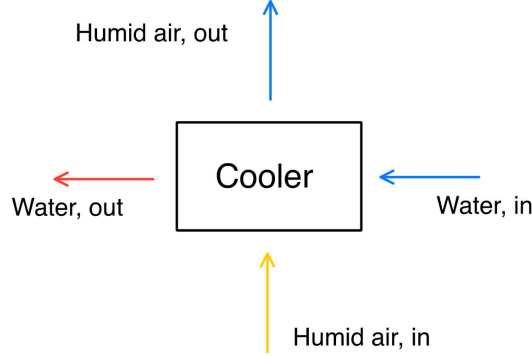


Figure 4.1: Equivalent representation of cooler system

From Part 0:

$$\dot{G}_{\text{humid air,inlet}} = \dot{G}_{\text{humid air,outlet}} = 3.75 \frac{\text{kg}}{\text{s}}$$

Instead, the mass flow rate of the water entering the cooler is obtained from an enthalpy balance around the cooler. The temperature values are known:

$$\dot{G}_{\text{water,inlet}} = \dot{G}_{\text{water,outlet}} = 2.25 \frac{\text{kg}}{\text{s}}$$

$$T_{\text{humid air,inlet}} = 25.4 \text{ }^{\circ}\text{C} \quad T_{\text{humid air,outlet}} = 13 \text{ }^{\circ}\text{C}$$

$$T_{\text{water,inlet}} = 7 \text{ }^{\circ}\text{C} \quad T_{\text{water,outlet}} = 12 \text{ }^{\circ}\text{C}$$

To evaluate the exergy content of water, the assumption of incompressible fluid is taken into account. In this case:

$$b_t = h - h_0 - T_0 \cdot (s - s_0) = c_p \cdot (T - T_0) - T_0 \cdot c_p \cdot \ln \left(\frac{T}{T_0} \right)$$

$$b_{\text{water,inlet}} = 4.186 \cdot (280 - 303.65) - 303.65 \cdot 4.186 \cdot \ln \left(\frac{7 + 273}{30.5 + 273} \right) = 4.06 \frac{\text{kJ}}{\text{kg}}$$

$$b_{\text{water,outlet}} = 4.186 \cdot (285 - 303.65) - 303.65 \cdot 4.186 \cdot \ln \left(\frac{12 + 273}{30.5 + 273} \right) = 2.50 \frac{\text{kJ}}{\text{kg}}$$

For humid air, the exergy content can be evaluated using the following formula:

$$b_t = (c_{p,a} + x \cdot c_{p,v}) \cdot T_0 \cdot \left(\frac{T}{T_0} - 1 - \ln \left(\frac{T}{T_0} \right) \right) + (1 + x) \cdot R^a \cdot T_0 \cdot \ln \left(\frac{P}{P_0} \right) \\ + R^a \cdot T_0 \cdot \left((1 + x) \ln \left(\frac{1 + \bar{x}_0}{1 + \bar{x}} \right) + x \ln \left(\frac{\bar{x}}{\bar{x}_0} \right) \right)$$

Constants:

$$c_{p,a} = 1 \frac{\text{kJ}}{\text{kg} \cdot \text{C}} \quad c_{p,v} = 1.9 \frac{\text{kJ}}{\text{kg} \cdot \text{C}} \quad R^a = 0.288 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

Assuming:

$$P = 1 \text{ atm}$$

$$x_{\text{inlet}} = 0.02 \frac{\text{kg}_{\text{water}}}{\text{kg}_{\text{dry_air}}} \quad x_{\text{outlet}} = 0.009 \frac{\text{kg}_{\text{water}}}{\text{kg}_{\text{dry_air}}}$$

$$b_{\text{humid air,inlet}} = (1 + 0.02 \cdot 1.9) \cdot (30.5 + 273) \cdot \left(\frac{28.3 + 273}{30.5 + 273} - 1 - \ln \left(\frac{28.3 + 273}{30.5 + 273} \right) \right) \\ + (1 + 0.037) \cdot 0.288 \cdot (30.5 + 273) \cdot \ln \left(\frac{1}{1} \right) \\ + 0.288 \cdot (30.5 + 273) \cdot \left((1 + 0.037) \ln \left(\frac{1 + 0.041}{1 + 0.026} \right) + 0.037 \ln \left(\frac{0.037}{0.041} \right) \right) \\ = 0.22 \frac{\text{kJ}}{\text{kg}}$$

$$b_{\text{humid air,outlet}} = (1 + 0.009 \cdot 1.9) \cdot (30.5 + 273) \cdot \left(\frac{13 + 273}{30.5 + 273} - 1 - \ln \left(\frac{13 + 273}{30.5 + 273} \right) \right) \\ + (1 + 0.015) \cdot 0.288 \cdot (30.5 + 273) \cdot \ln \left(\frac{1}{1} \right) \\ + 0.288 \cdot (30.5 + 273) \cdot \left((1 + 0.015) \ln \left(\frac{1 + 0.041}{1 + 0.026} \right) + 0.015 \ln \left(\frac{0.015}{0.041} \right) \right) \\ = 0.72 \frac{\text{kJ}}{\text{kg}}$$

Figure 4.2: Table 10: Outlet and inlet conditions of humid air and water through cooler system

To write the exergy equation, the cooler system is assumed as an adiabatic system with no heat dispersion and no external technical power. The exergy equation can be written in the subsequent way:

$$\dot{G}_{\text{water}} \cdot (b_{\text{out}} - b_{\text{in}}) + \dot{G}_{\text{humid,air}} \cdot (b_{\text{out}} - b_{\text{in}}) + T_0 \cdot \dot{\psi}_{\text{irr}} = 0$$

$$\text{Resources} = \dot{G}_{\text{water}} \cdot (b_{\text{in}} - b_{\text{out}}) = 3.52 \text{ kW}$$

$$\text{Products} = \dot{G}_{\text{humid,air}} \cdot (b_{\text{out}} - b_{\text{in}}) = 1.87 \text{ kW}$$

$$\text{Irreversibility} = \frac{\text{Resources} - \text{Products}}{T_0} = 0.0054 \frac{\text{kW}}{\text{K}}$$

$$\text{Efficiency} = \frac{\text{Products}}{\text{Resource}} = 0.531$$

Post heater

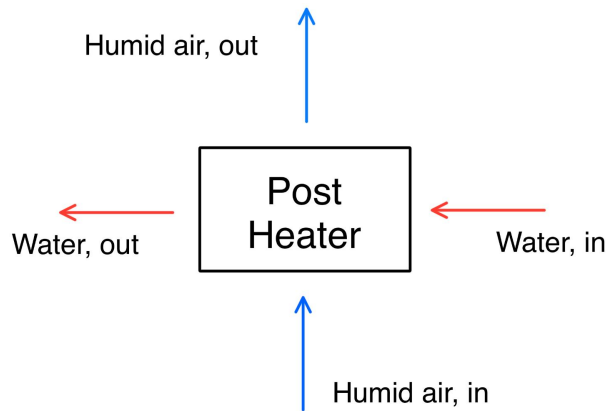


Figure 4.3: Simplified draw of post heater

Post heater is very similar to cooler. For this reason, the entire calculation is not reported here.

Figure 4.4: Table 11: Conditions of air and water through the post heater

The same assumption of an adiabatic system made before is considered true here. The exergy equation can be written in the subsequent way:

$$\dot{G}_{\text{water}} \cdot (b_{\text{out}} - b_{\text{in}}) + \dot{G}_{\text{humid,air}} \cdot (b_{\text{out}} - b_{\text{in}}) + T_0 \cdot \dot{\psi}_{\text{irr}} = 0$$

In general, the efficiency can always be written as

$$\text{Efficiency} = 1 - \frac{T_0 \cdot \dot{\psi}_{\text{irr}}}{\text{Resource}}$$

In order to write it, the resources and products terms are defined as follows:

$$\text{Resources} = \dot{G}_{\text{humid,air}} \cdot (b_{\text{in}} - b_{\text{out}}) = 0.442 \text{ kW}$$

$$\text{Products} = \dot{G}_{\text{water}} \cdot (b_{\text{out}} - b_{\text{in}}) = 0.217 \text{ kW}$$

$$\text{Irreversibility} = \frac{\text{Resources} - \text{Products}}{T_0} = 0.0007 \frac{\text{kW}}{\text{K}}$$

$$\text{Efficiency} = \frac{\text{Products}}{\text{Resource}} = 0.49$$

Whole HVAC system

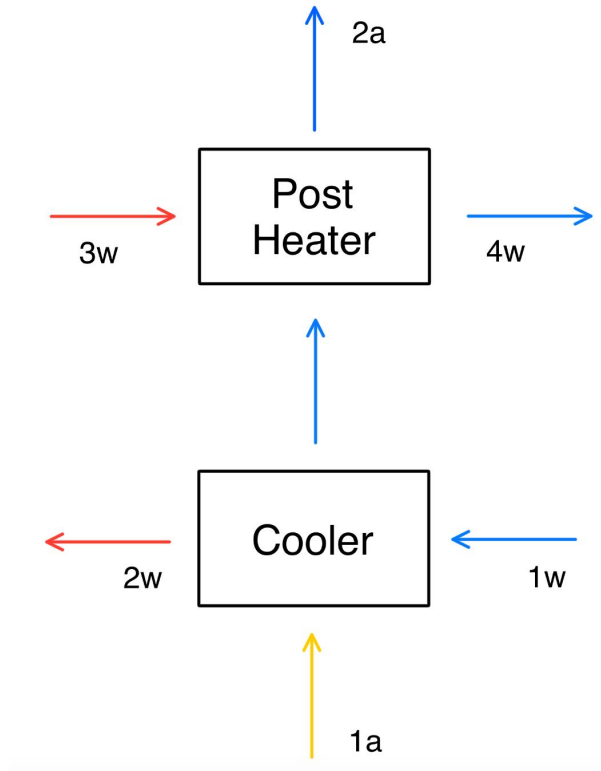


Figure 4.5: Simplified representation of whole HVAC system

The entire control volume is an adiabatic system with no exchange of technical power. Therefore, in the exergy equation there are only the terms related to mass flow rate and irreversibility. For the whole HVAC system, to identify the resources and the product is trickier. Since T_0 is higher than our circuit temperatures. The system would naturally move towards T_0 . So if the temperature of one of the liquid is decreased from inlet to outlet, it means the system was moved 'against' the natural movement, so it is a product. Instead, if the liquid in the heat exchanged increases its temperature, it means

that this liquid 'took the heat' of the other liquid, so it cooled it down, so it was a resource to cool another liquid down. Following this logic, we can define also product and resources for the whole HVAC system.

$$\dot{G}_{\text{air}} \cdot (b_{2a} - b_{1a}) + \dot{G}_{\text{water,cooler}} \cdot (b_{1w} - b_{2w}) + \dot{G}_{\text{water,post heater}} \cdot (b_{4w} - b_{3w}) + T_0 \cdot \dot{\psi}_{\text{irr}} = 0$$

- Humid air: Since the temperature of humid air from 1a to 2a is decreased (from 298.6K to 288K), it means that it is a product of a cooling process.
- Cooler water: The temperature of the water in the cooler is increased, therefore the cooler is a resource (cooling down humid air).
- Post-heater water: The temperature of the water in the post heater is decreased, so it is a product.

$$\text{Resources} = \dot{G}_{\text{water,cooler}} \cdot (b_{1w} - b_{2w}) = 3.52 \frac{\text{kJ}}{\text{kg}}$$

$$\text{Products} = \dot{G}_{\text{air}} \cdot (b_{2a} - b_{1a}) + \dot{G}_{\text{water,post heater}} \cdot (b_{4w} - b_{3w}) = 1.64 \frac{\text{kJ}}{\text{kg}}$$

$$\text{Irreversibility} = \frac{\text{Resources} - \text{Products}}{T_0} = 0.0062 \frac{\text{kJ}}{\text{K}}$$

$$\text{Efficiency} = \frac{\text{Products}}{\text{Resource}} = 0.467$$

An important observation is that:

$$\sum_i^{\text{components}} \dot{\psi}_{\text{irr},i} = \dot{\psi}_{\text{irr,HVAC}}$$

The sum of the irreversibility of cooler and post heater is equal to the irreversibility of the entire HVAC system.

We can start sumitting the data of the R134a Cycle:

Points	1	2	3	4
T [°C]	-3.00	45.50	35.50	-3.00
p [bar]	2.62	9.00	9.00	2.62
h [kJ/kg]	395.41	427.25	249.49	249.49
s [kJ/kgK]	1.72	1.74	1.17	1.18
x	1	gas	0	0.268 (gas+L)

Table 4.1: Refrigeration cycle

4.1.2 Refrigeration cycle

Evaporator

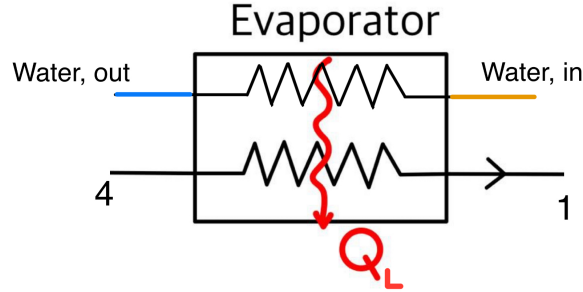


Figure 4.6: Evaporator scheme

The details of points 1 and 4 of R134a are reported in the previous table. Regarding waters, we know from the catalogue the following data:

$$T_{\text{water,in}} = 12^\circ\text{C}$$

$$T_{\text{water,out}} = 7^\circ\text{C}$$

$$b_t = h - h_0 - T_0 \cdot (s - s_0) = cp \cdot (T - T_0) - T_0 \cdot cp \cdot \ln \frac{T}{T_0}$$

$$b_{\text{water,inlet}} = 4.186 \cdot (285,15 - 303.65) - 303.65 \cdot 4.186 \cdot \ln \left(\frac{285,15}{303.65} \right) = 2,458 \frac{\text{kJ}}{\text{kg}}$$

$$b_{\text{water,outlet}} = 4.186 \cdot (280.15 - 303.65) - 303.65 \cdot 4.186 \cdot \ln \left(\frac{280.15}{303.65} \right) = 4,012 \frac{\text{kJ}}{\text{kg}}$$

$$b_{\text{R134a,outlet}} = h - h_0 - T_0(s - s_0) = 395,41 - 429.14 - (303.65)(1,72 - 1.9155) = 25,63 \frac{\text{kJ}}{\text{kg}}$$

$$b_{\text{R134a,inlett}} = 249,49 - 429.14 - (303.65)(1,18 - 1.9155) = 43,68 \frac{\text{kJ}}{\text{kg}}$$

	R134a		WATER	
	COLD-IN	COLD-OUT	HOT-IN	HOT-OUT
Flow rate [kg s^{-1}]	0,93	0,93	6,50	6,50
Temperature [$^\circ\text{C}$]	-3	-3	12	7
Exergy [kJ kg^{-1}]	25,634	43,685	2,458	4,012

Table 4.2: Thermodynamic properties at inlet and outlet for R134a and water

The system is considered adiabatic because the heat released by the water is absorbed by the R134a; therefore, there is no heat exchange with the surroundings.

$$\dot{G}_{\text{water}} \cdot (b_{\text{out}} - b_{\text{in}}) + \dot{G}_{\text{R134a}} \cdot (b_1 - b_4) + T_0 \cdot \dot{\psi}_{\text{irr}} = 0$$

$$\text{Products} = \dot{G}_{\text{water}} \cdot (b_{\text{out}} - b_{\text{in}}) = 10,108 \text{ kW}$$

$$\text{Resources} = \dot{G}_{\text{R134a}} \cdot (b_4 - b_1) = 16,824 \text{ kW}$$

$$\text{Irreversibility} = \frac{\text{Resources} - \text{Products}}{T_0} = 0,0221 \frac{\text{kW}}{\text{K}}$$

$$\text{Efficiency} = \frac{\text{Products}}{\text{Resources}} = 0,608$$

Condenser

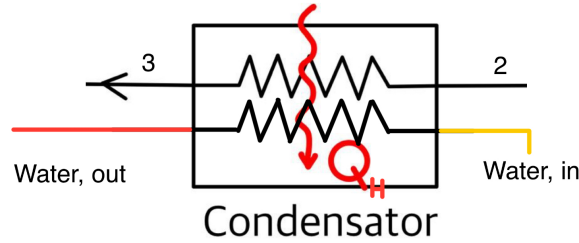


Figure 4.7: Condenser scheme

	R134a		WATER	
	HOT-IN	HOT-OUT	COLD-IN	COLD-OUT
Flow rate [kg s^{-1}]	0,93	0,93	1	1
Temperature [$^{\circ}\text{C}$]	45,50	35,50	21,06	22,85
Exergy [kJ kg^{-1}]	51,40	46,72	0,627	0,410

Table 4.3: Thermodynamic properties at inlet and outlet for R134a and water in the condenser

In this case, the condenser is not considered adiabatic, unlike the evaporator, because not all the heat from the R134a is exchanged with the water; part of it is also transferred to the surrounding environment. This environment is assumed to be at the dead state temperature T_0 . Since the heat is exchanged at $T = T_0$, the Carnot factor becomes zero:

$$\varphi = 1 - \frac{T_0}{T} = 1 - \frac{T_0}{T_0} = 0$$

As a result, even though the thermal power exchanged is different from zero, the related term in the exergy balance equation vanishes, because the useful work potential (exergy) associated with this heat transfer is null.

$$\dot{G}_{\text{water}} \cdot (b_{\text{out}} - b_{\text{in}}) + \dot{G}_{\text{R134a}} \cdot (b_3 - b_2) + T_0 \cdot \dot{\psi}_{\text{irr}} = 0$$

$$\text{Products} = \dot{G}_{\text{water}} \cdot (b_{\text{in}} - b_{\text{out}}) = 0,217 \text{ kW}$$

$$\text{Resources} = \dot{G}_{\text{R134a}} \cdot (b_3 - b_2) = 4,361 \text{ kW}$$

$$\text{Irreversibility} = \frac{\text{Resources} - \text{Products}}{T_0} = 0,0136 \frac{\text{kW}}{\text{K}}$$

$$\text{Efficiency} = \frac{\text{Products}}{\text{Resources}} = 0,05$$

Compressor

	R134a	
	IN-1	OUT-2
Flow rate [kg s ⁻¹]	0,93	0,93
Temperature [°C]	-3,00	45,50
Enthalpy [kJ kg ⁻¹]	395,41	427,25
Entropy [kJ kg ⁻¹ K ⁻¹]	1,72	1,74
Exergy [kJ kg ⁻¹]	25,63	51,40

Table 4.4: Thermodynamic properties at inlet and outlet for R134a in the compressor

In this case, there is no thermal contribution from the water; only the energy and entropy flows associated with the inlet and outlet of the R134a are considered. However, the work input W_t required to operate the compressor must be included in the balance with a negative sign.

$$-W_t = \dot{G}_{\text{R134a}} \cdot (b_2 - b_1) + T_0 \cdot \dot{\psi}_{\text{irr}}$$

$$\text{Products} = \dot{G}_{\text{R134a}} \cdot (b_2 - b_1) = 24,015 \text{ kW}$$

$$\text{Resources} = -W_t = 29,990 \text{ kW}$$

$$\text{Irreversibility} = \frac{\text{Resources} - \text{Products}}{T_0} = 0,02 \frac{\text{kW}}{\text{K}}$$

$$\text{Efficiency} = \frac{\text{Products}}{\text{Resources}} = 0,801$$

Throttling valve

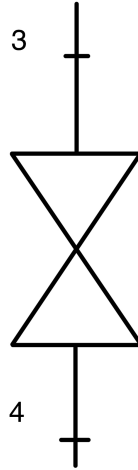


Figure 4.8: Throttling valve scheme

	R134a	
	IN-3	OUT-4
Flow rate [kg s ⁻¹]	0,93	0,93
Temperature [°C]	35,50	-3,00
Enthalpy [kJ kg ⁻¹]	249,49	249,49
Entropy [kJ kg ⁻¹ K ⁻¹]	1,17	1,18
Exergy [kJ kg ⁻¹]	46,72	43,68

Table 4.5: Thermodynamic properties at inlet and outlet for R134a in the valve

$$\dot{G}_{\text{R134a}} \cdot (b_4 - b_3) + T_0 \cdot \dot{\psi}_{\text{irr}} = 0$$

$$\text{Products} = \dot{G}_{\text{R134a}} \cdot b_4 = 40,714 \text{ kW}$$

$$\text{Resources} = \dot{G}_{\text{R134a}} \cdot b_3 = 43,544 \text{ kW}$$

$$\text{Irreversibility} = \frac{\text{Resources} - \text{Products}}{T_0} = 0,01 \frac{\text{kW}}{\text{K}}$$

$$\text{Efficiency} = \frac{\text{Products}}{\text{Resources}} = 0,935$$

Whole refrigerating cycle

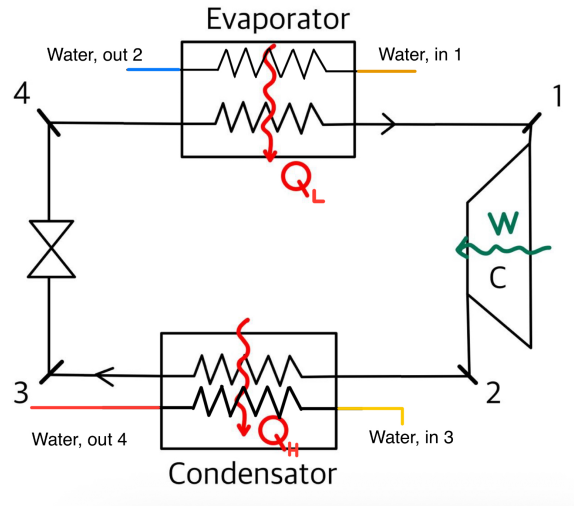


Figure 4.9: Whole cycle scheme

As reported in the condenser part, the whole cycle is not an adiabatic process, but the heat exchange is nil.

$$-W_t = \dot{G}_{\text{water, evaporator}} \cdot (b_{2w} - b_{1w}) + \dot{G}_{\text{water, condenser}} \cdot (b_{4w} - b_{3w}) + T_0 \cdot \dot{\psi}_{\text{irr}}$$

$$\text{Resources} = -W_t = 29,99 \text{ kW}$$

$$\text{Products} = \dot{G}_{\text{water, evaporator}} \cdot (b_{2w} - b_{1w}) + \dot{G}_{\text{water, condenser}} \cdot (b_{4w} - b_{3w}) = 10,325 \text{ kW}$$

$$\text{Irreversibility} = \frac{\text{Resources} - \text{Products}}{T_0} = \frac{29,99 - 10,325}{T_0} = 0,0648 \frac{\text{kW}}{\text{K}}$$

$$\text{Efficiency} = \frac{\text{Products}}{\text{Resources}} = \frac{10,325}{29,99} = 0,344$$

Also in this case, similarly to the HVAC system, the total irreversibility of the refrigerating cycle is given by the sum of the irreversibilities of each individual component:

$$\sum_i \dot{\psi}_{\text{irr},i} = \dot{\psi}_{\text{irr, refrigerating cycle}} = 0,065 \frac{\text{kW}}{\text{K}}$$

That is, the irreversibility of the overall refrigerating cycle is the result of the contributions from all system components.

Conclusions

The table below reports the exergetic efficiencies of each component, along with the corresponding exergy destructions, calculated as:

$$\dot{\psi}_{\text{irr},i} \cdot T_0$$

Table 4.6: Exergy efficiency and destruction of system components

	Evap	Cond	Lam valve	Cooler	Post heater	Compressor	Total
Component exergy efficiency	60.08%	4.97%	93.50%	46.25%	49.04%	80.08%	–
Exergy destruction [kJ/kg]	6.716	4.578	2.830	2.328	0.251	5.975	22.677
Exergy destruction [%]	29.61%	20.19%	12.48%	10.26%	1.10%	26.35%	100.00%

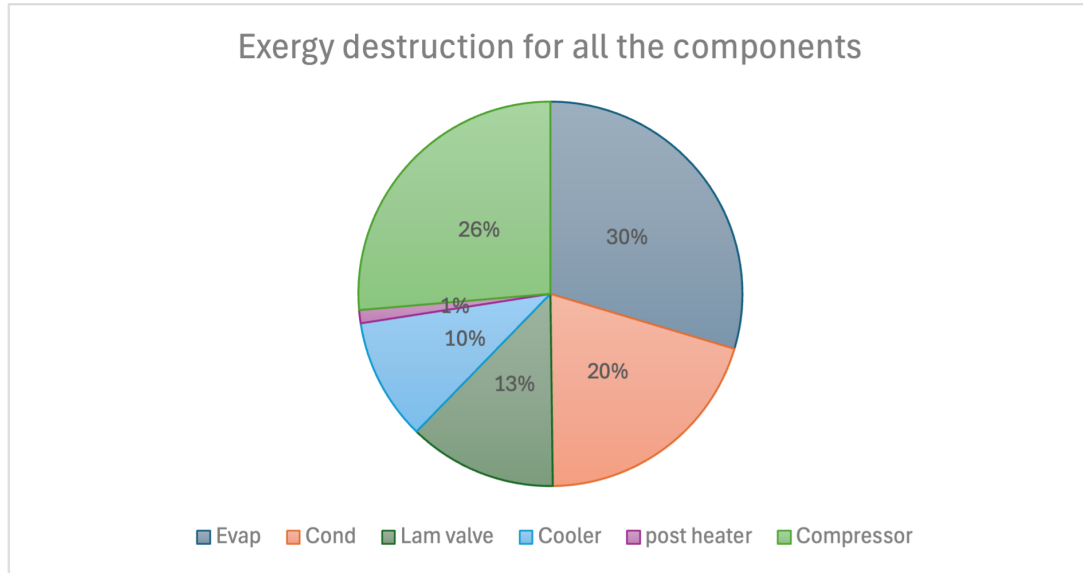


Figure 4.10: Exergy destruction diagram

4.2 Winter case

4.2.1 Definition of the dead state

The chosen dead state conditions are chosen because the heat treatment unit is located in the laboratory on the ground floor and are reported in the following tables:

$T_0[^\circ C]$	20
$p_0[Pa]$	101324
x_0^*	0.002576
$MM_{\text{dry air}}$	28.1
MM_{vap}	18
x_0	0.00165
$h_0[kJ/kg]$	419.96
$s_0[kJ/kg * K]$	1884.71

Table 4.7: Dead state conditions

where, $x_0^* = x_0 * \frac{MM_{\text{dry air}}}{MM_{\text{vap}}}$ and $x_0 = 0.00165 \frac{kg_{H_2O}}{kg_{\text{dry air}}}$

$T_0[^\circ C]$	20
$p_0[Pa]$	101324
x_0	0.002576

Table 4.8: Dead state humid air

Coolpack was used to get the properties listed in table 4.9

$T_0[^\circ C]$	20
$p_0[Pa]$	101324
$h_{R134a0}[kJ/kg]$	419.96
$s_{R134a0}[kJ/kg * K]$	1884.71

Table 4.9: Dead state R134a0

To evaluate the exergy of water, just as in the summer case, it is assumed to have an incompressible fluid. The main properties are listed in table 4.10

T_0	20
$p_0[Pa]$	101324
$h_{w,0}[kJ/kg]$	84.013
$s_{w,0}[kJ/kg * K]$	0.296

Table 4.10: Dead state water

4.2.2 HVAC

Pre-heater

For the water:

WATER	IN	OUT
$T_w[K]$	338.15	328.15
$T_0[K]$	293.15	293.15
$c_{p,water}$	4.183	4.183

Table 4.11: Caption

$$b_{w, i} = c_{p,w} \cdot (T_{w,i} - T_0) - (T_0 \cdot c_{p,w} \cdot \ln(\frac{T_{w,i}}{T_0})) = 13.1207 \frac{kJ}{kg}$$

$$b_{w, o} = c_{p,w} \cdot (T_{w,o} - T_0) - (T_0 \cdot c_{p,w} \cdot \ln(\frac{T_{w,o}}{T_0})) = 13.1207 J = 8.1011 \frac{kJ}{kg}$$

For the air, the main data are listed in the table below:

AIR	IN	OUT
$c_{p,a}$	1	1
$c_{p,v}$	1.9	1.9
T_0	293.15	293.15
T_a	$T_{a,m} = 287$	$T_{a,1} = 305$
x_*	0.0089	0.0089
R_a^*	0.288	0.288
P	101324	101324
P_0	101324	101324
x_0^*	0.00258	0.00258

Table 4.12

where:

$$x_* = x_m * \frac{MM_{\text{dry air}}}{MM_{\text{vap}}} \text{ and } x_m = 0.00571 \frac{kg_{H_2O}}{kg_{\text{dry air}}}$$

$$\begin{aligned}
b_{t,a,in} &= (c_{p,a} + x_0 \cdot c_{p-v}) \cdot T_0 \cdot \left(\frac{T_{a,m}}{T_0} - 1 - \ln \left(\frac{T_{a,m}}{T_0} \right) + (1 + x_*) \cdot R_a^* \cdot T_0 \cdot \ln \left(\frac{P}{P_0} \right) \right. \\
&\quad \left. + R_a^* \cdot T_0 \cdot \left((1 + x_*) \cdot \ln \left(\frac{1 + x_0^*}{1 + x_0} \right) + x_* \cdot \ln \left(\frac{x_*}{x_0^*} \right) \right) \right) \\
&= 1.079 \text{ kJ/kg}
\end{aligned}$$

$$\begin{aligned}
b_{t,a,out} &= (c_{p,a} + x_0 \cdot c_{p-v}) \cdot T_0 \cdot \left(\frac{T_{a,1}}{T_0} - 1 - \ln \left(\frac{T_{a,1}}{T_0} \right) + (1 + x_*) \cdot R_a^* \cdot T_0 \cdot \ln \left(\frac{P}{P_0} \right) \right. \\
&\quad \left. + R_a^* \cdot T_0 \cdot \left((1 + x_*) \cdot \ln \left(\frac{1 + x_0^*}{1 + x_0} \right) + x_* \cdot \ln \left(\frac{x_*}{x_0^*} \right) \right) \right) \\
&= 1.25 \text{ kJ/kg}
\end{aligned}$$

$$G_{w,in} = \frac{G_{\text{humid air}} \cdot c_{p,a} (T_{\text{humid air, out}} - T_{\text{humid air, in}})}{(T_{\text{water, in}} - T_{\text{water, out}}) \cdot c_{p,\text{water}}} = 2.55 \text{ kg/s}$$

$$G_{\text{humid air, in}} = 5.89 \text{ kg/s}$$

$$Resources = G_{\text{water, in}} \cdot (b_{\text{water, in}} - b_{\text{water, out}}) = 12.81 \text{ kW}$$

$$Products = G_{\text{humid air, in}} \cdot (b_{\text{t, air, in}} - b_{\text{t, air, out}}) = 1.03 \text{ kW}$$

$$\psi_{irr} = \frac{Resources - Products}{T_0} = 0.04 \frac{\text{kW}}{\text{K}}$$

$$Efficiency = \frac{Products}{Resources} = 8.04\%$$

Humidifier

$G_{\text{steam, in}}$	0.00452
$T_{\text{steam, in}}$	373.15
h_{steam}	2675
$G_{\text{humid air, in}}$	5.89
$T_{\text{humid air, in}}$	305.15
$T_{\text{humid air, out}}$	305.15

Table 4.13: Main parameters for the following calculations

$h_{\text{steam, in}}$	2675.5
h_0	83.91
s	7.355
s_0	0.2967

Table 4.14: Parameters to calculate $b_{\text{steam, in}}$

$$b_{\text{steam, in}} = (h_{\text{steam, in}} - h_0) - T_0 \cdot (s - s_0) = 522.4494 \text{ kJ/kg}$$

AIR	IN	OUT
$c_{p,a}$	1	1
$c_{p,v}$	1.9	1.9
T_0	293.15	293.15
T_a	$T_{a,1} = 293.15$	$T_{a,2} = 294.15$
x_*	0.0089	0.01012
R_a^*	0.288	0.288
P	101324	101324
P_0	101324	101324
x_0^*	0.00258	0.00258
x_0	0.00165	0.00165
	$x_1 = 0.00571$	$x_2 = 0.00648$

Table 4.15

$$\begin{aligned}
b_{t,a,in} &= (c_{p,a} + x_0 \cdot c_{p-v}) \cdot T_0 \cdot \left(\frac{T_{a,m}}{T_0} - 1 - \ln \left(\frac{T_{a,1}}{T_0} \right) + (1 + x_*) \cdot R_a^* \cdot T_0 \cdot \ln \left(\frac{P}{P_0} \right) \right. \\
&\quad \left. + R_a^* \cdot T_0 \cdot \left((1 + x_*) \cdot \ln \left(\frac{1 + x_0^*}{1 + x_0} \right) + x_* \cdot \ln \left(\frac{x_*}{x_0^*} \right) \right) \right) \\
&= .25 \text{ kJ/kg}
\end{aligned}$$

$$\begin{aligned}
b_{t,a,out} &= (c_{p,a} + x_0 \cdot c_{p-v}) \cdot T_0 \cdot \left(\frac{T_{a,1}}{T_0} - 1 - \ln \left(\frac{T_{a,1}}{T_0} \right) + (1 + x_*) \cdot R_a^* \cdot T_0 \cdot \ln \left(\frac{P}{P_0} \right) \right. \\
&\quad \left. + R_a^* \cdot T_0 \cdot \left((1 + x_*) \cdot \ln \left(\frac{1 + x_0^*}{1 + x_0} \right) + x_* \cdot \ln \left(\frac{x_*}{x_0^*} \right) \right) \right) \\
&= 1.49 \text{ kJ/kg}
\end{aligned}$$

$$Resources = G_{\text{steam, in}} \cdot b_{\text{steam, in}} + G_{\text{humid air, in}} \cdot b_{\text{t, air, in}} = 9.75 \text{ kW}$$

$$Products = G_{\text{humid air, in}} \cdot b_{\text{t, air, out}} = 8.76 \text{ kW}$$

$$\psi_{\text{irr}} = \frac{Resources - Products}{T_0} = 0.0034 \frac{\text{kW}}{\text{K}}$$

$$Efficiency = \frac{Products}{Resources} = 89.88\%$$

4.2.3 Total HVAC Plant

To further extend the exergy analysis, the entire HVAC system can be considered as a single control volume. The exergy of the products is then evaluated

$$Products = G_{\text{humid, air, in}} = b_{\text{t, air, out}} - b_{\text{t, air, in}} = 4.335 \text{ kJ/kg}$$

The resources consist of the mass flow rates of water supplied to the heat exchangers and the mass flow rate of steam generated by the humidifier.

$$Resources = G_{\text{water, in}} \cdot (b_{\text{water, in}} - b_{\text{water, out}}) + G_{\text{steam, in}} \cdot b_{\text{steam, in}} + G_{\text{water, in}} \cdot (b_{\text{water, in}} - b_{\text{water, out}}) = 27.62 \text{ kJ/kg}$$

The irreversibility:

$$Irreversibility = \frac{Resources - Products}{T_0} = 0.0435 \frac{\text{kJ}}{\text{K}}$$

Finally, the efficiency:

$$Efficiency = \frac{Products}{Resources} = 15.86\%$$

4.2.4 Refrigeration cycle

For the R134a Cycle:

Points	1	2	3	4
T [°C]	-18.00	84.40	75.00	-18.00
p [bar]	1.45	23.62	23.62	1.45
h [kJ/kg]	386.33	444.42	313.27	313.25
s [kJ/kgK]	1.73	1.75	1.36	1.45
x	1	gas	0	0.651 (gas + L)

Table 4.16: Refrigeration cycle winter

Evaporator

The details of points 1 and 4 of R134a are reported in the previous table. Regarding waters, we know from the catalogue the following data:

$$T_{\text{R134, in}} = 255.15 \text{ K}$$

$$T_{\text{R134, out}} = 255.15 \text{ K}$$

$$T_{\text{amb, winter}} = 265.15 \text{ K}$$

$$b_{R134,4} = (h_4 - h_0) - T_0 \cdot (s_4 - s_0) = 21.605 kJ/kg$$

$$b_{R134,1} = (h_1 - h_0) - T_0 \cdot (s_1 - s_0) = 10.55 kJ/kg$$

The same considerations that were made for the summer case are assumed here as well

$$\text{Resources} = -Q_{\text{evaporator}} \left(1 - \frac{T_0}{T_{\text{amb,water}}}\right) = 41.00 kJ/kg$$

$$\text{Products} = G_{R134a} \cdot (b_{R14,1} - b_{R14,4}) = -33.87 kJ/kg$$

$$\text{Irreversibility} = \frac{\text{Resources} - \text{Products}}{T_0} = 0.255 \frac{kW}{K}$$

$$\text{Efficiency} = \frac{-\phi_{\text{evaporator}} \left(1 - \frac{T_0}{T_{\text{amb,wint}}}\right)}{G_{R134} \cdot (b_{R134,4} - b_{R134,1})} = 82.63\%$$

Condenser

	R134a		WATER	
	IN	OUT	IN	OUT
Flow rate [kg s ⁻¹]	2.96	2.96	5.171	5.171
Temperature [K]	357.55	348.15	328.15	338.15
Exergy [kJ kg ⁻¹]	62.78	47.72	8.101	13.12

Table 4.17: Thermodynamic properties at inlet and outlet for R134a and water in the condenser

$$\dot{G}_{\text{water}} \cdot (b_{\text{out}} - b_{\text{in}}) + \dot{G}_{R134a} \cdot (b_3 - b_2) + T_0 \cdot \dot{\psi}_{\text{irr}} = 0 \quad (4.1)$$

$$\text{Products} = \dot{G}_{\text{water}} \cdot (b_{\text{out}} - b_{\text{in}}) = 25.96 kW \quad (4.2)$$

$$\text{Resources} = \dot{G}_{R134a} \cdot (b_{\text{in}} - b_{\text{out}}) = 44.58 kW \quad (4.3)$$

$$\dot{\psi}_{\text{irr}} = \frac{\text{Resources} - \text{Products}}{T_0} = 0.0635 kW/K \quad (4.4)$$

$$\text{Efficiency} = \frac{\text{Products}}{\text{Resources}} = 58.22\% \quad (4.5)$$

Compressor The value of the flow rate is the same of the condenser. The exergies are

obtained in the following way:

$$b_1 = (h_1 - h_0) - T_0 \cdot (s_1 - s_0) = 10.55 kJ/kg$$

$$b_1 = (h_2 - h_0) - T_0 \cdot (s_2 - s_0) = 62.78 kJ/kg$$

$$-W_t = \dot{G}_{R134a} \cdot (b_2 - b_1) + T_0 \cdot \dot{\psi}_{irr} = 171.88 kW$$

$$\text{Products} = \dot{G}_{R134a} \cdot (b_2 - b_1) = 154.592 kW$$

$$\text{Resources} = -W_t = 171.88 kJ/kg$$

$$\text{Irreversibility} = \frac{\text{Resources} - \text{Products}}{T_0} = 0.059 \frac{kJ}{K}$$

$$\text{Efficiency} = \frac{\text{Products}}{\text{Resources}} = 89.94\%$$

Lamination valve

$$\dot{G}_{R134a} \cdot (b_4 - b_3) + T_0 \cdot \dot{\psi}_{irr} = 0$$

$$\text{Products} = \dot{G}_{R134a} \cdot b_4 = 63.95 kW$$

$$\text{Resources} = \dot{G}_{R134a} \cdot b_3 = 141.24 kW$$

$$\text{Irreversibility} = \frac{\text{Resources} - \text{Products}}{T_0} = 0.264 \frac{kJ}{K}$$

$$\text{Efficiency} = \frac{\text{Products}}{\text{Resources}} = 45.28\%$$

Whole refrigerating cycle

As reported in the condenser part, the whole cycle is not an adiabatic process, but the heat exchange is nil.

$$-W_t = \dot{G}_{\text{water, evaporator}} \cdot (b_{2w} - b_{1w}) + \dot{G}_{\text{water, condenser}} \cdot (b_{4w} - b_{3w}) + T_0 \cdot \dot{\psi}_{irr}$$

$$\text{Resources} = \text{Resources}_{\text{evaporator}} + \text{Resources}_{\text{compressor}} = 219 kW$$

$$\text{Products} = \dot{G}_{\text{water, in}} \cdot (b_{\text{water, in}} - b_{\text{water, out}}) = 25.96 kW$$

$$\text{Irreversibility} = \frac{\text{Resources} - \text{Products}}{T_0} = 0.658 \frac{\text{kW}}{\text{K}}$$

$$\text{Efficiency} = \frac{\text{Products}}{\text{Resources}} = \frac{10,325}{29,99} = 11.86\%$$

Also in this case, similarly to the HVAC system, the total irreversibility of the refrigerating cycle is given by the sum of the irreversibilities of each individual component:

$$\sum_i \dot{\psi}_{\text{irr},i} = \dot{\psi}_{\text{irr, refrigerating cycle}} = 0,658 \frac{\text{kW}}{\text{K}}$$

That is, the irreversibility of the overall refrigerating cycle is the result of the contributions from all system components.

Conclusions

The table below reports the exergetic efficiencies of each component, along with the corresponding exergy destructions, calculated as:

$$\dot{\psi}_{\text{irr},i} \cdot T_0$$

Table 4.18: Exergy efficiency and destruction of system components

	Evap	Cond	Lam valve	Compressor	Pre Heater	Humidifier	Total
Component exergy efficiency	82.63%	56.24%	45.28%%	89.94%	8.04%	89.88%	
Exergy destruction [kJ/kg]	74.86	20.20	80.00	17.90	11.78	0.99	205.72
Exergy destruction [%]	36.39%	9.82%	38.89%	8.70%	5.72%	0.48%	100%

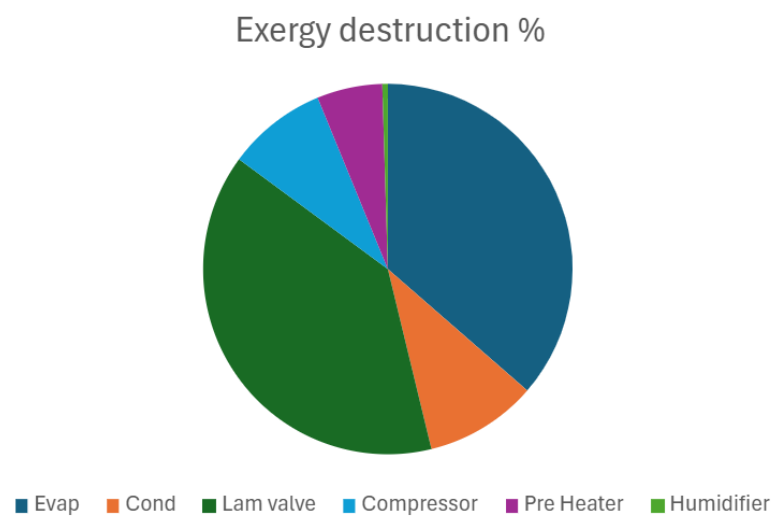


Figure 4.11: Exergy destruction in winter

Chapter 5

Thermal design of components

This part of the project is related to the sizing of the air-and-water heat exchanges in the fan coil for both summer and winter conditions. As requested the heat exchanger features a crossflow configuration. The aim is evaluating the length of the coil from the thermal power transferred and the evaluation of the coil total thermal resistance. Tube sizes has been chosen as OD 3/8 inches ($\varnothing 9.52 \text{ mm}$) with thickness 0.76 mm found as common practice values. The material selected for the tube is copper due to its high thermal conductivity and, indeed, common use in coils for heat exchangers. The thermal resistance of the coil is retrieved, with reference to Figure 5.1 and Table 5.1 for quantities, as follows:

$$R_{tot} = R_i + R_{wall} + R_o = \frac{1}{h_i A_i} + \frac{\ln \frac{D_o}{D_i}}{2\pi k L} + \frac{1}{h_o A_o} \quad (5.1)$$

D_i	8	mm
A_i	50.3	mm^2
D_o	9.52	mm
A_o	71.2	mm^2
λ_{Cu}	398	$\frac{W}{m \cdot K}$

Table 5.1: Tube dimensions and material properties

The convective heat transfer coefficients of water inside the tube h_i and that of air h_o are estimated by means of the dimensional analysis described by the Buckingham theorem, passing through the dimensionless groups Re, Nu and Pr.

The reference fan coil is the one used for the definition of temperatures in the gas refrigeration cycle of Part 0, namely the FWF-03DF by DAIKIN (see Datasheet in the Appendix), whose main specifications for the "medium" operating condition are listed in the following tables:

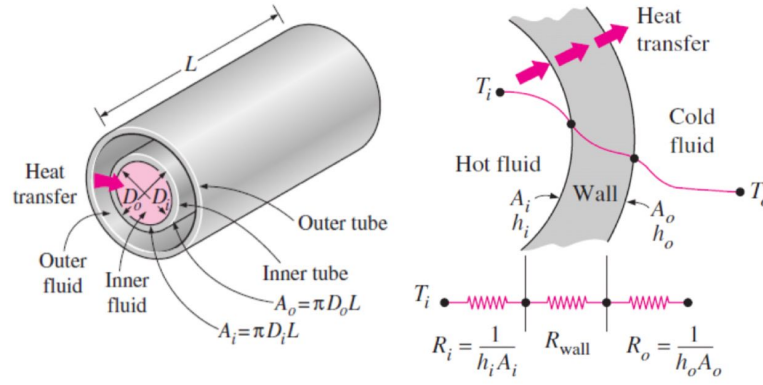


Figure 5.1: Tube thermal resistance

$T_{water,in}$	$70^{\circ}C$
$\dot{G}_{air}$	$463 \frac{l}{hr}$
$\dot{G}_{water}$	$310 \frac{l}{hr}$

Table 5.2: Fan coil properties in heating mode

$T_{water,in}$	$7^{\circ}C$
$\dot{G}_{air}$	$463 \frac{l}{hr}$
$\dot{G}_{water}$	$476 \frac{l}{hr}$

Table 5.3: Fan coil properties in cooling mode

5.1 Summer condition

5.1.1 ε -NTU method

The ε -NTU method (also effectiveness-NTU method) is used to obtain the overall heat transfer coefficient UA . It can be used when the mass flow rates and inlet stream temperatures are known, but the outlet are not.

The first step is to compute the heat capacity rates for both the air and water:

$$\left\{ \begin{array}{l} C_{air} = \dot{G}_{air} \cdot c_{p,air} \\ \quad = 0.150 \cdot 1006 \\ \quad = 151.12 \frac{W}{K} \\ \\ C_{water} = \dot{G}_{water} \cdot c_{p,water} \\ \quad = 0.132 \cdot 4203 \\ \quad = 555.65 \frac{W}{K} \end{array} \right.$$

Next we can identify which one is the minimum and which is the maximum:

$$\begin{aligned} C_{min} &= C_{air} = 151.12 \frac{W}{K} \\ C_{max} &= C_{water} = 555.65 \frac{W}{K} \end{aligned}$$

From here we can evaluate the ratio of the two:

$$c = \frac{C_{max}}{C_{min}} = 0.272 \quad (5.2)$$

This value will be useful when computing the number of transfer units (NTU). To compute that value we also need the effectiveness, which is the ratio of the actual heat being transferred, and the maximum heat which can be transferred. The actual heat being transferred in summer is:

$$\Phi = 1630 \text{ W} \quad (5.3)$$

To get this value, we computed the thermal load of the most critical office following the same procedure as the one done in Part 0. We will size the heat exchanger only for this office and then consider all others to be the same.

The maximum heat which can be transferred is computed as:

$$\Phi_{max} = C_{min} \Delta T_{max}$$

We use C_{min} since that is the limiting heat capacity rate. The highest temperature difference is that of the two inlets, in summer it is the temperature at the inlet of the air minus that of the water:

$$\Delta T = T_{air,in} - T_{water,in} = 25^\circ C - 7^\circ C = 18^\circ C \quad (= 18 \text{ K}) \quad (5.4)$$

Therefore we can now compute the maximum heat transferable:

$$\Phi_{max} = 151.12 \cdot 18 = 2720 \text{ W} \quad (5.5)$$

Therefore the effectiveness is:

$$\begin{aligned} \varepsilon &= \frac{\Phi}{\Phi_{max}} \\ &= \frac{1630}{2720} \\ &= 0.599 \end{aligned} \quad (5.6)$$

By considering a configuration with C_{max} mixed and C_{min} unmixed we can use the equation highlighted in Figure 5.2 to compute the NTU

TABLE B	NTU relation
NTU relations for heat exchangers $NTU = UA_s/C_{min}$ and $c = C_{min}/C_{max} = (\dot{m}C_p)_{min}/(\dot{m}C_p)_{max}$ (Kays and London, Ref. 5.)	
Heat exchanger type	NTU relation
1 <i>Double-pipe:</i> Parallel-flow	$NTU = -\frac{\ln [1 - \varepsilon(1 + c)]}{1 + c}$
Counter-flow	$NTU = \frac{1}{c - 1} \ln \left(\frac{\varepsilon - 1}{\varepsilon c - 1} \right)$
2 <i>Shell and tube:</i> One-shell pass 2, 4, . . . tube passes	$NTU = -\frac{1}{\sqrt{1 + c^2}} \ln \left(\frac{2/\varepsilon - 1 - c - \sqrt{1 + c^2}}{2/\varepsilon - 1 - c + \sqrt{1 + c^2}} \right)$
3 <i>Cross-flow (single-pass)</i> C_{max} mixed, C_{min} unmixed	$NTU = -\ln \left[1 + \frac{\ln (1 - \varepsilon c)}{c} \right]$
C_{min} mixed, C_{max} unmixed	$NTU = -\frac{\ln [c \ln (1 - \varepsilon) + 1]}{c}$
4 <i>All heat exchangers</i> with $c = 0$	$NTU = -\ln(1 - \varepsilon)$

Figure 5.2: NTU Formulas

$$\begin{aligned}
 NTU &= -\ln \left[1 + \frac{\ln (1 + \varepsilon c)}{c} \right] \\
 &= -\ln \left[1 + \frac{\ln(1 + 0.599 \cdot 0.272)}{0.272} \right] \\
 &= 1.06
 \end{aligned} \tag{5.7}$$

To obtain this value it is also possible to use some charts available, however, we decided to only use this formula.

We now have the NTU , it is important to recall, as it will be useful later, an important equation that correlated NTU to the total heat transfer coefficient:

$$NTU \cdot C_{min} = UA \tag{5.8}$$

5.1.2 Convective heat transfer coefficient

The convective coefficients (h_i and h_o) are determined using correlations based on dimensionless numbers. Specifically, they are calculated through the Nusselt number (Nu), which relates to the convective heat transfer. The Nusselt number itself is obtained from empirical correlations involving other dimensionless parameters, such as the Reynolds and Prandtl numbers.

$$Re = \frac{vD}{\nu}$$

$$Pr = \frac{\mu c_p}{k}$$

$$Nu = f(Re, Pr) = \frac{hd}{\lambda}$$

$$h = \frac{Nu\lambda}{d}$$

Internal convective heat transfer coefficient h_i

First, the water flow rate must be considered, as it is specified in the datasheet of the chosen fan coil. The internal diameter of the tube is known, so the speed of the fluid can be calculated using the following equation:

$$v_{water} = \frac{G_{water}}{A_i} = 2.63 m/s$$

Using the aforementioned equations, the Reynolds and Prandtl numbers are obtained:

$$Re = 1.47 \times 10^4$$

$$Pr = 10.39$$

Next, the friction factor is calculated using the following empirical relation:

$$f = (1.58 \cdot \ln(Re) - 3.28)^{-2} = 7.08 \times 10^{-3}$$

Then, the Nusselt number is computed using another empirical equation, and the internal convective heat transfer coefficient is determined:

$$Nu = \frac{\frac{f}{2} \cdot Re \cdot Pr}{1 + 8.7 \cdot \left(\frac{f}{2}\right)^{1/2} \cdot (Pr - 1)} = 92.53$$

$$h_i = \frac{\lambda \cdot Nu}{d} = 6673 \frac{W}{m^2 K}$$

External convective heat transfer coefficient h_o

Outside the pipes, air flows freely in a perpendicular direction with respect to that of the water. In order to evaluate h_o , it is necessary to provide some dimensions of the geometry of the heat exchanger: the following table contains a preliminary estimation of the internal shape of the fancoil.

The lateral surface A_{lat} is given by the space between two consecutive coils, multiplied

Number of rows (NT)	5	/
Pipes per row (NL)	6	/
ST	55	mm
SL	30	mm
Length of tube (L)	500	mm

Table 5.4: Preliminary dimensions of the heat exchanger

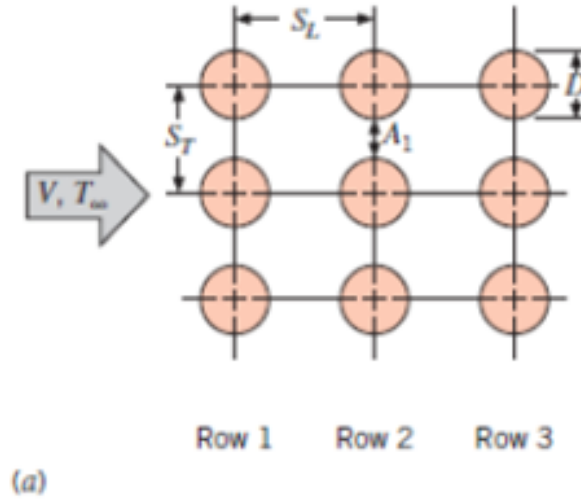


Figure 5.3

by the number of lateral intervals (eight using this configuration).

$$\begin{aligned}
 A_{lat} &= (S_T - D) \cdot L \cdot (NT - 1) \\
 &= (55 - 9.52) \cdot 500 \cdot (5 - 1) \cdot 10^{-6} \\
 &= 0.091 \, m^2
 \end{aligned} \tag{5.9}$$

The velocity of the air can now be calculated:

$$v_{air} = \frac{G_{air}}{A_{lat}} = 1.41 \, m/s$$

The Zhukauskas approach was used to calculate the Reynolds number:

$$v_{max} = \frac{ST}{ST_D} v_{air} = 1.71 \, m/s$$

where $ST_D = ST - D = 35.48 \, mm$

$$Re_{max} = \frac{v_{max} D}{\nu_{air}} = 1049$$

Finally, since the pipes were assumed to be in an ‘aligned’ configuration, we can

calculate the Nusselt number using the following empirical equation:

$$\text{Nu} = C \cdot \text{Re}_{\max}^m \cdot \text{Pr}^{0.36} \cdot \left(\frac{\text{Pr}}{\text{Pr}_s} \right)^{1/4} = 6.15$$

TABLE 7.5 Constants of Equation 7.50 for the tube bank in cross flow [16]

Configuration	$\text{Re}_{D,\max}$	C_1	m
Aligned	$10-10^2$	0.80	0.40
Staggered	$10-10^2$	0.90	0.40
Aligned	10^2-10^3	Approximate as a single (isolated) cylinder	
Staggered	10^2-10^3		
Aligned	$10^3-2 \times 10^5$	0.27	0.63
$(S_T/S_L > 0.7)^a$			
Staggered	$10^3-2 \times 10^5$	$0.35(S_T/S_L)^{1/5}$	0.60
$(S_T/S_L < 2)$			
Staggered	$10^3-2 \times 10^5$	0.40	0.60
$(S_T/S_L > 2)$			
Aligned	$2 \times 10^5-2 \times 10^6$	0.021	0.84
Staggered	$2 \times 10^5-2 \times 10^6$	0.022	0.84

^aFor $S_T/S_L < 0.7$, heat transfer is inefficient and aligned tubes should not be used.

TABLE 7.6 Correction factor C_2 of Equation 7.51 for $N_L < 20$ ($\text{Re}_{D,\max} \geq 10^3$) [16]

N_L	1	2	3	4	5	7	10	13	16
Aligned	0.70	0.80	0.86	0.90	0.92	0.95	0.97	0.98	0.99
Staggered	0.64	0.76	0.84	0.89	0.92	0.95	0.97	0.98	0.99

Figure 5.4: Nusselt and Nusselt correction factor

where: $C = 0.27$ and $m = 0.63$.

Furthermore, since we have $N_L=6$ we must multiply Nu by a correction factor $C_2 = 0.93$:

$$\text{Nu}_{\text{corr}} = 5.72 \quad (5.10)$$

The convective heat transfer coefficient for the air around the pipes is then calculated as:

$$h_o = \frac{\lambda \cdot \text{Nu}_{\text{corr}}}{d} = 364.16 \frac{\text{W}}{\text{m}^2 \text{K}}$$

5.1.3 Radiator size

Once the two convective heat transfer coefficients are computed, by following the electrical resistance analogy seen in Figure 5.1 we can compute U_{tot} :

$$\begin{aligned} U_{\text{tot}} &= \left(\frac{1}{h_i} \cdot \frac{d_o}{d_i} + \frac{\ln\left(\frac{d_o}{d_i}\right)}{2\lambda} \cdot d_o + \frac{1}{h_o} \right)^{-1} \\ &= 364 \frac{\text{W}}{\text{m}^2 \text{K}} \end{aligned} \quad (5.11)$$

With this computed, we can compute the needed external surface of the pipes:

$$\begin{aligned} A_{tot} &= \frac{NTU \cdot C_{min}}{U_{tot}} \\ &= 0.44 \text{ m} \end{aligned} \quad (5.12)$$

Next we compute the total tube length needed to have this external area:

$$L_{tot} = \frac{A_{tot}}{\pi D} = 14.72 \text{ m} \quad (5.13)$$

Finally, by imposing the length of the heat exchanger as 0.5 m we can compute the number of tubes necessary:

$$N_{tot} = \frac{L_{tot}}{L} = 29.45 \text{ tubes} \quad (5.14)$$

Rounding this up gives us the initial value that we estimated of 30 tubes.

If this value would have been different in theory we would have had to consider a new configuration, recompute the external convective heat transfer coefficient and re-perform all the steps until the previous one in an iterative manner, until we converge to a number of tubes. However, we can note that if we consider the rows fixed and equal to 5, we can add columns without changing any of the parameters that follow the choice of the configuration, therefore, for this case and all the following cases, the number of tubes selected will be a multiple of 5.

5.1.4 Fins

The table below provides the calculated values of the internal and external convective heat transfer coefficients. It is evident from these results that there is a significant difference between them, particularly with the external surface exhibiting a much lower heat transfer coefficient.

Coefficient	Value
h_i	$6673.69 \frac{W}{m^2 \cdot K}$
h_o	$364.16 \frac{W}{m^2 \cdot K}$

Table 5.5: Convective heat transfer coefficients without the use of fins

From these values, it can be inferred that the external surface acts as a thermal bottleneck, limiting the total heat transfer rate. The overall heat transfer coefficient, in this case, is calculated to be $U_{tot} = 374 \frac{W}{m^2 \cdot K}$.

To optimize the heat exchange process and minimize the required tube length and number, finned cross-flow coils can be employed. By keeping the same system

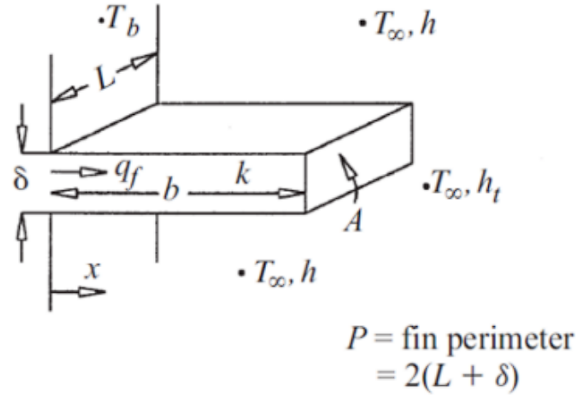


Figure 5.5: Schematic view of a single fin

configuration as the base case (without fins), we can enhance performance while maintaining the same flow rate.

Although adding fins increases the available heat exchange surface, not all of this added area contributes equally. Therefore, we introduce a fin efficiency factor to account for the reduced effectiveness. This efficiency, under the assumption of an adiabatic tip, is:

$$\eta = \frac{\tanh(m \cdot b)}{m \cdot b} = 0.971 \quad (5.15)$$

Where:

- $b = 6 \text{ mm}$ is the height of the fin.
- $\delta = 1 \text{ mm}$ is the assumed fin thickness.

The parameter m is defined as:

$$m = \sqrt{\frac{2h_o}{\delta k}} = 43.49 \left[\frac{1}{\text{m}} \right] \quad (5.16)$$

This efficiency is valid for a single fin. To extend the analysis to all fins, an overall efficiency term must be introduced:

$$\eta_{ov} \cdot S = S_b + \eta S_f \quad (5.17)$$

Where the components are defined as:

- S_b : non-finned surface area $= \pi D_o(L - \delta \cdot N_f) = 0.01 \text{ m}^2$
- S_f : total finned surface $= (S_T \cdot S_L - \pi D_o^2/4) \cdot 2N_f = 0.32 \text{ m}^2$
- S : total effective heat exchange area $= S_b + S_f = 0.33 \text{ m}^2$
- N_f : number of fins $= 220$ fins (assumed value)

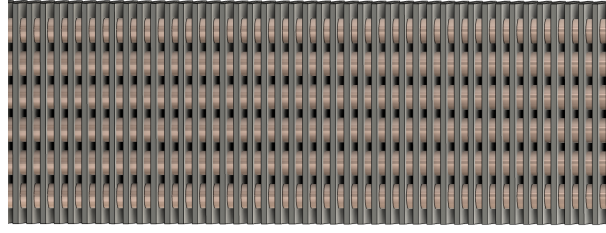


Figure 5.6: Fin and pipe configuration

The resulting overall fin efficiency is:

$$\eta_{ov} = 0.972$$

The internal heat transfer coefficient h_i remains unchanged. To evaluate the external coefficient h_o under the new configuration, we must account for increased air velocity due to reduced cross-sectional flow area.

The new average air velocity is:

$$v_{air} = \frac{G_{air}}{A_{lat}} = \frac{G_{air}}{(S_T - D) \cdot N_{vertical-pipes} \cdot (L - \delta \cdot N_f)} = 2.52 \text{ m/s} \quad (5.18)$$

Using the Zhukauskas method again, we compute the peak velocity and Reynolds number:

$$v_{max} = \frac{S_T}{S_T - D} \cdot v_{air} = 3.05 \text{ m/s} \quad (5.19)$$

$$Re_{max} = \frac{v_{max} \cdot D}{\nu} = 1872.86 \quad (5.20)$$

The Nusselt number is estimated as:

$$Nu_{corrected} = C_2 \cdot C \cdot Re_{max}^m \cdot Pr^{0.36} \cdot \left(\frac{Pr}{Pr_s} \right)^{1/4} = 8.24 \quad (5.21)$$

where $m=0.63$, $C_2 = 0.93$ and $C=0.27$.

From this, we determine the updated external heat transfer coefficient:

$$h_o = \frac{\lambda \cdot Nu}{D} = 524.72 \frac{W}{m^2 \cdot K} \quad (5.22)$$

where $\lambda = 385 \frac{W}{m \cdot K}$

This change leads to a slightly improved total thermal transmittance, now calculated as:

$$U_{total,i} = \left[\frac{1}{h_i} + \frac{\ln\left(\frac{D}{D_i}\right)}{2k} + \frac{D_i}{\eta_{ov} \cdot h_o \cdot D} \right]^{-1} = 555.77 \frac{W}{m^2 \cdot K} \quad (5.23)$$

Representative values of the overall heat transfer coefficients in heat exchangers

Type of heat exchanger	U , $W/m^2 \cdot ^\circ C^*$
Water-to-water	850–1700
Water-to-oil	100–350
Water-to-gasoline or kerosene	300–1000
Feedwater heaters	1000–8500
Steam-to-light fuel oil	200–400
Steam-to-heavy fuel oil	50–200
Steam condenser	1000–6000
Freon condenser (water cooled)	300–1000
Ammonia condenser (water cooled)	800–1400
Alcohol condensers (water cooled)	250–700
Gas-to-gas	10–40
Water-to-air in finned tubes (water in tubes)	30–60 [†] 400–850 [†]
Steam-to-air in finned tubes (steam in tubes)	30–300 [†] 400–4000 [‡]

Figure 5.7: Typical range of heat transfer coefficients for various exchanger types

Although fins improve thermal efficiency, they also contribute to increased resistance and potential pumping penalties. Their use should therefore be optimized, not maximized. The observed heat transfer coefficients are consistent with typical ranges found in the literature, as shown in the figure below

The overall heat transfer coefficients with and without fins are summarized below:

U_{tot}	$(\frac{W}{m^2 \cdot K})$
Without fins	374
With fins	555.77

Table 5.6: Comparison of U_{tot} with and without fins

A_i	0.29 m^2
L_{tot}	9.65 m
N_{tubes}	20

Table 5.7: Tube requirement in summer conditions using fins

We can see that we reduced the number of tubes from 30 to 20 by implementing the use of fins.

5.2 Winter condition

5.2.1 ε -NTU method

Analogously to what reported in Section 5.1.1 for the summer condition, the computation of the overall heat transfer coefficient UA starts with determining the heat capacity rates for air and water:

$$\begin{cases} C_{air} &= \dot{G}_{air} \cdot c_{p,air} \\ &= 0.155 \cdot 1006 \\ &= 155.78 \frac{W}{K} \\ C_{water} &= \dot{G}_{water} \cdot c_{p,water} \\ &= 0.084 \cdot 4188 \\ &= 353.61 \frac{W}{K} \end{cases}$$

The maximum and minimum value are then defined:

$$\begin{aligned} C_{min} &= C_{air} = 155.78 \frac{W}{K} \\ C_{max} &= C_{water} = 353.61 \frac{W}{K} \end{aligned}$$

and the ratio of the two is obtained:

$$c = \frac{C_{max}}{C_{min}} = 0.441 \quad (5.24)$$

Together with c , to compute the number of transfer units (NTU), we also need the effectiveness, which is the ratio of the actual heat being transferred, and the maximum heat which can be transferred. For winter case, the former is:

$$\Phi = 3176 \text{ W} \quad (5.25)$$

Also in this case, it has been obtained from the thermal load of the most critical office following the same procedure of Part 0. We will size the heat exchanger only for this office and then consider all others to be the same.

The maximum heat which can be transferred, instead, is computed as:

$$\Phi_{max} = C_{min} \Delta T_{max}$$

Again, we use C_{min} since that is the limiting heat capacity rate. The highest

temperature difference is that of the two inlets, which is:

$$\Delta T = T_{water,in} - T_{air,in} = 65^\circ C - 20^\circ C = 45^\circ C \quad (= 45 K) \quad (5.26)$$

Therefore the maximum heat transferable is:

$$\Phi_{max} = 155.78 \cdot 45 = 7010 W \quad (5.27)$$

Therefore the effectiveness is:

$$\begin{aligned} \varepsilon &= \frac{\Phi}{\Phi_{max}} \\ &= \frac{3176}{7010} \\ &= 0.453 \end{aligned} \quad (5.28)$$

By considering the same configuration with C_{max} mixed and C_{min} unmixed, we can use again Equation 5.7 to compute the NTU:

$$\begin{aligned} NTU &= -\ln \left[1 + \frac{\ln(1 + \varepsilon c)}{c} \right] \\ &= -\ln \left[1 + \frac{\ln(1 + 0.453 \cdot 0.441)}{0.441} \right] \\ &= 0.704 \end{aligned} \quad (5.29)$$

5.2.2 Convective heat transfer coefficient

The procedure is the same of the one used for the summer case, and so is the geometry of the fancoil. The following tables contain the values of the physical parameters necessary to evaluate h_i and h_o .

v_{water}	1.71	m/s
Re	$3.1 \cdot 10^4$	/
Pr	2.77	/
f	$5.86 \cdot 10^{-3}$	/
Nu	137.39	/
h_i	11248.98	$\frac{W}{m^2 K}$

Table 5.8: Water

v_{air}	1.41	m/s
v_{max}	1.71	m/s
Re_{max}	1081	/
Nu	6.26	/
Nu_{corr}	5.83	/
h_o	365.97	$\frac{W}{m^2K}$

Table 5.9: Air

5.2.3 Radiator size

Analogously to Section 5.1.3, we can compute U_{tot} :

$$U_{tot,ext} = \left(\frac{1}{h_i} \cdot \frac{d_o}{d_i} + \frac{\ln\left(\frac{d_o}{d_i}\right)}{2\lambda} \cdot d_o + \frac{1}{h_o} \right)^{-1} = 366 \frac{W}{m^2K} \quad (5.30)$$

Now, we can compute the needed external surface of the pipes simply as:

$$\begin{aligned} A_{tot} &= \frac{NTU \cdot C_{min}}{U_{tot}} \\ &= 0.30 \, m \end{aligned} \quad (5.31)$$

and the total tube length needed to have this external area is:

$$L_{tot} = \frac{A_{tot}}{\pi D} = 10.02 \, m \quad (5.32)$$

which, always considering the length of the heat exchanger as 0.5 m, yields to a minimum number of tubes of:

$$N_{tot} = \frac{L_{tot}}{L} = 20.04 \, \text{tubes} \quad (5.33)$$

In this case we would need at least 21 tubes, but as said before, to keep the same 5 tubes per row, we would decide for 25 tubes. Nevertheless, we can see that summer is the most critical as we would need 30 tubes (without fins). So out of the two configurations without fins we would need 30 tubes

5.2.4 Fins

The table below provides the calculated values of the internal and external convective heat transfer coefficients.

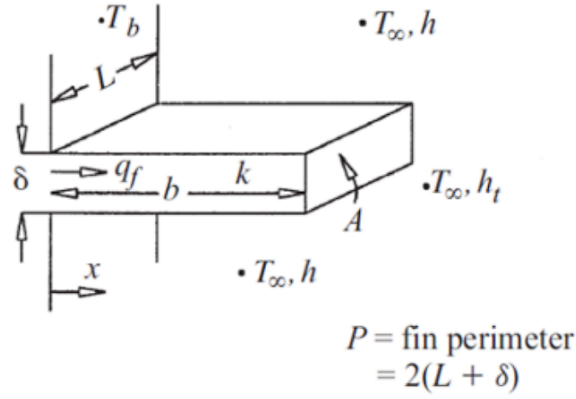


Figure 5.8: Schematic view of a single fin

Coefficient	Value
h_i	$11248.98 \frac{W}{m^2 \cdot K}$
h_o	$376.97 \frac{W}{m^2 \cdot K}$

Table 5.10: Convective heat transfer coefficients without the use of fins

The overall heat transfer coefficient, in this case, is calculated to be $U_{tot} = 366 \frac{W}{m^2 \cdot K}$. Although adding fins increases the available heat exchange surface, not all of this added area contributes equally. Therefore, we introduce a fin efficiency factor to account for the reduced effectiveness. This efficiency, under the assumption of an adiabatic tip, is:

$$\eta = \frac{\tanh(m \cdot b)}{m \cdot b} = 0.971 \quad (5.34)$$

Where:

- $b = 6 \text{ mm}$ is the height of the fin.
- $\delta = 1 \text{ mm}$ is the assumed fin thickness.

The parameter m is defined as:

$$m = \sqrt{\frac{2h}{\delta k}} = 43.6 \left[\frac{1}{m} \right] \quad (5.35)$$

This efficiency is valid for a single fin. To extend the analysis to all fins, an overall efficiency term must be introduced:

$$\eta_{ov} \cdot S = S_b + \eta S_f \quad (5.36)$$

Where the components are defined as:

- S_b : non-finned surface area $= \pi D_o(L - \delta \cdot N_f) = 0.01 \text{ m}^2$

- S_f : total finned surface = $(S_T \cdot S_L - \pi D_o^2/4) \cdot 2N_f = 0.32 \text{ m}^2$
- S : total effective heat exchange area = $S_b + S_f = 0.33 \text{ m}^2$
- N_f : number of fins = 220 fins (assumed value)

The resulting overall fin efficiency is:

$$\eta_{ov} = 0.972$$

The internal heat transfer coefficient h_i remains unchanged. To evaluate the external coefficient h_o under the new configuration, we must account for increased air velocity due to reduced cross-sectional flow area.

The new average air velocity is:

$$v_{air} = \frac{G_{air}}{A_{lat}} = \frac{G_{air}}{(S_T - D) \cdot N_{vertical-pipes} \cdot (L - \delta \cdot N_f)} = 2.52 \text{ m/s} \quad (5.37)$$

Using the Zhukauskas method again, we compute the peak velocity and Reynolds number:

$$v_{max} = \frac{S_T}{S_T - D} \cdot v_{air} = 3.05 \text{ m/s} \quad (5.38)$$

$$Re_{max} = \frac{v_{max} \cdot D}{\nu} = 1930.60 \quad (5.39)$$

The Nusselt number is estimated as:

$$Nu_{corrected} = C_2 \cdot C \cdot Re_{max}^m \cdot Pr^{0.36} \cdot \left(\frac{Pr}{Pr_s}\right)^{1/4} = 8.39 \quad (5.40)$$

where $m=0.63$, $C_2 = 0.93$ and $C=0.27$.

From this, we determine the updated external heat transfer coefficient:

$$h_o = \frac{\lambda \cdot Nu}{D} = 574.84 \frac{W}{m^2 \cdot K} \quad (5.41)$$

where $\lambda = 385 \frac{W}{m \cdot K}$

This change leads to a slightly improved total thermal transmittance, now calculated as:

$$U_{total,i} = \left[\frac{1}{h_i} + \frac{\ln\left(\frac{D}{D_i}\right)}{2k} + \frac{D_i}{\eta_{ov} \cdot h_o \cdot D} \right]^{-1} = 630.00 \frac{W}{m^2 \cdot K} \quad (5.42)$$

Although fins improve thermal efficiency, they also contribute to increased resistance and potential pumping penalties. Their use should therefore be optimized, not maximized. The observed heat transfer coefficients are consistent with typical ranges found in the literature, as shown in the figure below

Representative values of the overall heat transfer coefficients in heat exchangers

Type of heat exchanger	U , $W/m^2 \cdot ^\circ C^*$
Water-to-water	850–1700
Water-to-oil	100–350
Water-to-gasoline or kerosene	300–1000
Feedwater heaters	1000–8500
Steam-to-light fuel oil	200–400
Steam-to-heavy fuel oil	50–200
Steam condenser	1000–6000
Freon condenser (water cooled)	300–1000
Ammonia condenser (water cooled)	800–1400
Alcohol condensers (water cooled)	250–700
Gas-to-gas	10–40
Water-to-air in finned tubes (water in tubes)	30–60 [†] 400–850 [†]
Steam-to-air in finned tubes (steam in tubes)	30–300 [†] 400–4000 [‡]

Figure 5.9: Typical range of heat transfer coefficients for various exchanger types

The overall heat transfer coefficients with and without fins are summarized below:

U_{tot}	$(\frac{W}{m^2 \cdot K})$
Without fins	366
With fins	630

Table 5.11: Comparison of U_{tot} with and without fins

A_i	0.17 m ²
L_{tot}	5.85 m
N_{tubes}	12

Table 5.12: Tube requirement in summer conditions using fins

For the same logic as before, here we would round the number of tubes up to 15, a decrease of 10 from without the use of fins.

In any case, we can see that if we use fins the summer condition is still the most critical and would require at least 20 tubes in a 5x4 configuration.

Cassetta BLDC a protocollo aperto

**Motore del ventilatore BLDC per installazione a soffitto.
Mandata d'aria a 4 vie**

- La struttura compatta (570 mm di larghezza e altezza) ne consente il montaggio a filo nel controsoffitto senza che sia necessario tagliare i pannelli, integrandosi nei moduli architettonici standard
- Griglia di aspirazione dell'aria moderna in ABS o dal design ultrapiatto
- Affidabilità e robustezza in un design compatto
- Pompa di scarico condensa con prevalenza fino a 835 mm
- Ampia gamma di sistemi di controllo a protocollo aperto
- Disponibilità di valvole a 2 o 3 vie con attuatore ON-OFF premontato in fabbrica



FWF-D

Per maggiori dettagli e informazioni approfondite, scansionare i codici QR o fare clic sugli stessi.

Unità interna			FWF	02DF	02DT	03DT	03DF	04DF	04DT	05DT	05DF
Capacità di raffreddamento (condizioni standard)	Capacità totale 2 tubi	Alta	kW	-	2,00	3,00	-		4,07	5,10	-
		Media	kW	-	1,67	2,78	-		3,41	4,16	-
		Bassa	kW	-	1,30	2,37	-		2,65	2,93	-
	Capacità totale 4 tubi	Alta	kW	2,00	-	-	3,00	4,00	-		5,02
		Media	kW	1,71	-	-	2,77	3,33	-		4,00
		Bassa	kW	1,44	-	-	2,30	2,58	-		2,64
	Capacità sensibile 2 tubi	Alta	kW	-	1,76	2,31	-		3,01	3,88	-
		Media	kW	-	1,43	2,08	-		2,49	3,08	-
		Bassa	kW	-	1,09	1,75	-		1,91	2,11	-
	Capacità sensibile 4 tubi	Alta	kW	1,76	-	-	2,19	2,88	-		3,67
		Media	kW	1,46	-	-	1,99	2,33	-		2,88
		Bassa	kW	1,20	-	-	1,61	1,78	-		1,85
Capacità latente 2 tubi	Alta	kW	-	0,24	0,69	-		1,06	1,22	-	
	Capacità latente 4 tubi	Alta	kW	0,24	-	-	0,81	1,12	-		1,35
Capacità di riscaldamento (condizioni standard)	Capacità 2 tubi	Alta	kW	-	2,54	3,30	-		4,26	5,74	-
		Media	kW	-	2,05	2,96	-		3,48	4,34	-
		Bassa	kW	-	1,56	2,44	-		2,69	2,95	-
	Capacità 4 tubi	Alta	kW	3,31	-	-	4,15	4,59	-		5,64
		Media	kW	2,77	-	-	3,61	3,75	-		4,32
		Bassa	kW	2,23	-	-	2,78	2,90	-		2,99
Potenza assorbita		kW	0,017	0,018	0,019		0,024		0,045	0,047	
			kW	0,01			0,02		0,01	0,02	
FCEER				129	121	188	156	174	180	120	130
FCCOP				220	156	197	193	198	194	128	174
CLASSE FCEER				B		A	B				
CLASSE FCCOP				B	C	B				C	B
Dimensioni	Unità	AltezzaxLarghezzaxProfondità	mm	260x642x575							
Peso	Unità	kg	16,0	14,5	15,5	17,0		15,5		17,0	
Ventilatore	Tipo		Ventilatore turbo								
	Quantità		1								
	Portata d'aria	Alta	m³/ora	477	498	516	534	612	623	860	847
		Media	m³/ora	389	388	455	463	487	496	634	607
Bassa		m³/ora	301	278	363	356	361	369	408	367	
Livello di potenza sonora totale	Alta	dBA	41,0		42,0		44,0	48,0	47,0	54,0	56,0
	Media	dBA	37,0		39,0		40,0	43,0	41,0	46,0	48,0
	Bassa	dBA	34,0	33,0	36,0		38,0		36,0	39,0	40,0
Livello di pressione sonora	Alta	dBA	27,0		28,0		30,0	34,0	33,0	40,0	42,0
	Media	dBA	23,0		25,0		26,0	29,0	27,0	32,0	34,0
	Bassa	dBA	20,0	19,0	22,0		24,0		22,0	25,0	26,0
Portata acqua	Raffrescamento	Alta	l/ora	345	344	515	516	687	699	878	864
		Media	l/ora	294	286	477	476	573	586	716	687
		Bassa	l/ora	248	224	407	396	444	455	504	455
	Riscaldamento	Alta	l/ora	285	437	568	357	395	733	987	485
		Media	l/ora	238	353	508	310	322	599	747	371
		Bassa	l/ora	192	269	419	239	249	463	507	257
Collegamenti tubazioni	Condensa	DE	mm	VP20 (diam. esterno 26/diam. interno 20)							
Alimentazione	Fase / Frequenza / Tensione		Hz/V	1~/50/230							

Raffreddamento: aria 27 °CBS, 19 °CBU; acqua in ingresso 7 °C; acqua in uscita 12 °C | Riscaldamento: 2 tubi: aria 20 °CBS, 15 °CBU; acqua in ingresso 45 °C; acqua in uscita 40 °C | Riscaldamento: 4 tubi: aria 20 °CBS, 15 °CBU; acqua in ingresso 65 °C; acqua in uscita 55 °C | L'unità non è precaricata. È presente una carica residua minima correlata all'ispezione qualità presso la fabbrica | Le misurazioni del valore della portata d'aria sono effettuate alle condizioni 20 °C(BS) /15 °C(BU).

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